# Get the Most Out of Visual Studio and Xamarin for Mobile Developers

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Microsoft

# David Ortinau

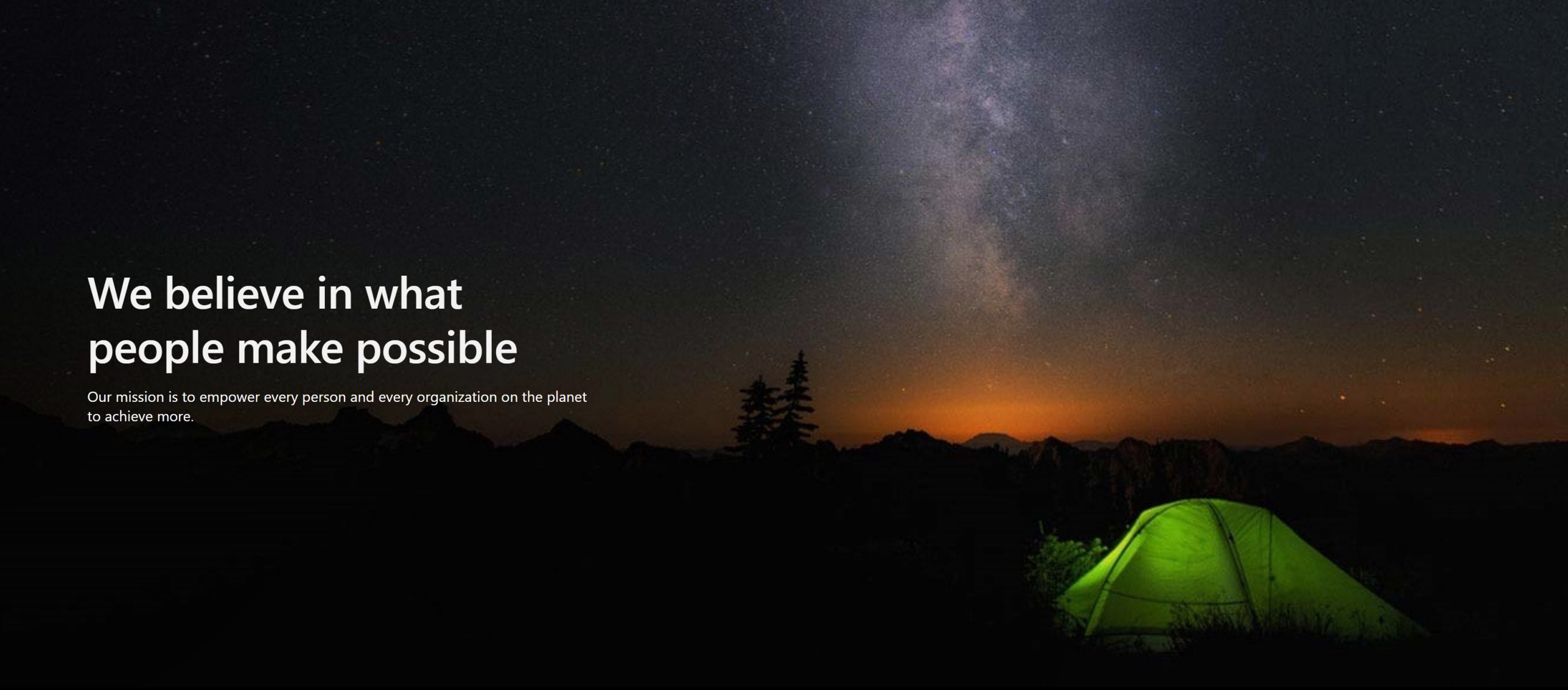
23 years web, interactive, mobile

Advertising, Marketing, Media
Healthcare

@davidortinau

david.ortinau@microsoft.com





# We believe in what people make possible

Our mission is to empower every person and every organization on the planet to achieve more.

# What we hear from you.

# 13% talk about build times:

“Bad integration with VS (slow builds , 'Clean and Rebuild' required too often, failed builds that require VS restarts, problems when VS is upgraded, etc.)”

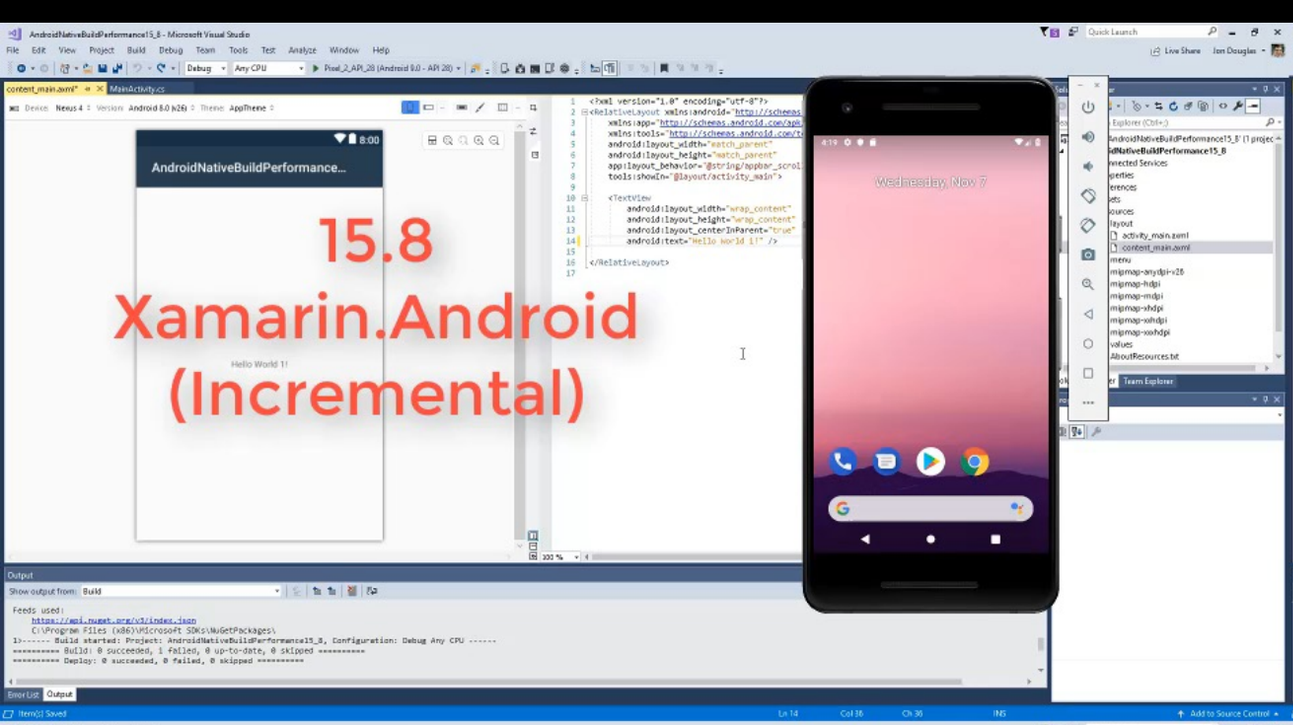
“Build times, random errors ( build , clean, rebuild”

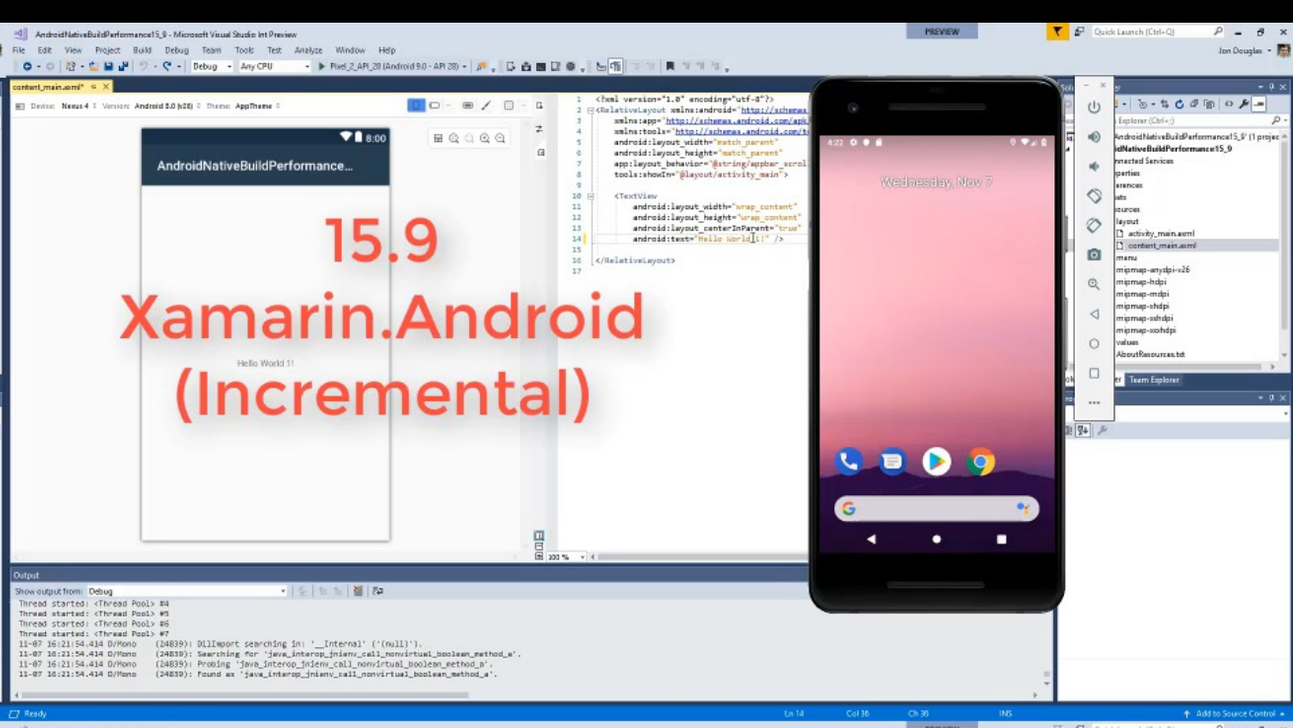
“debugging... I have a very powerful machine but it's still painfully slow to launch and build and often doesn't work first time”

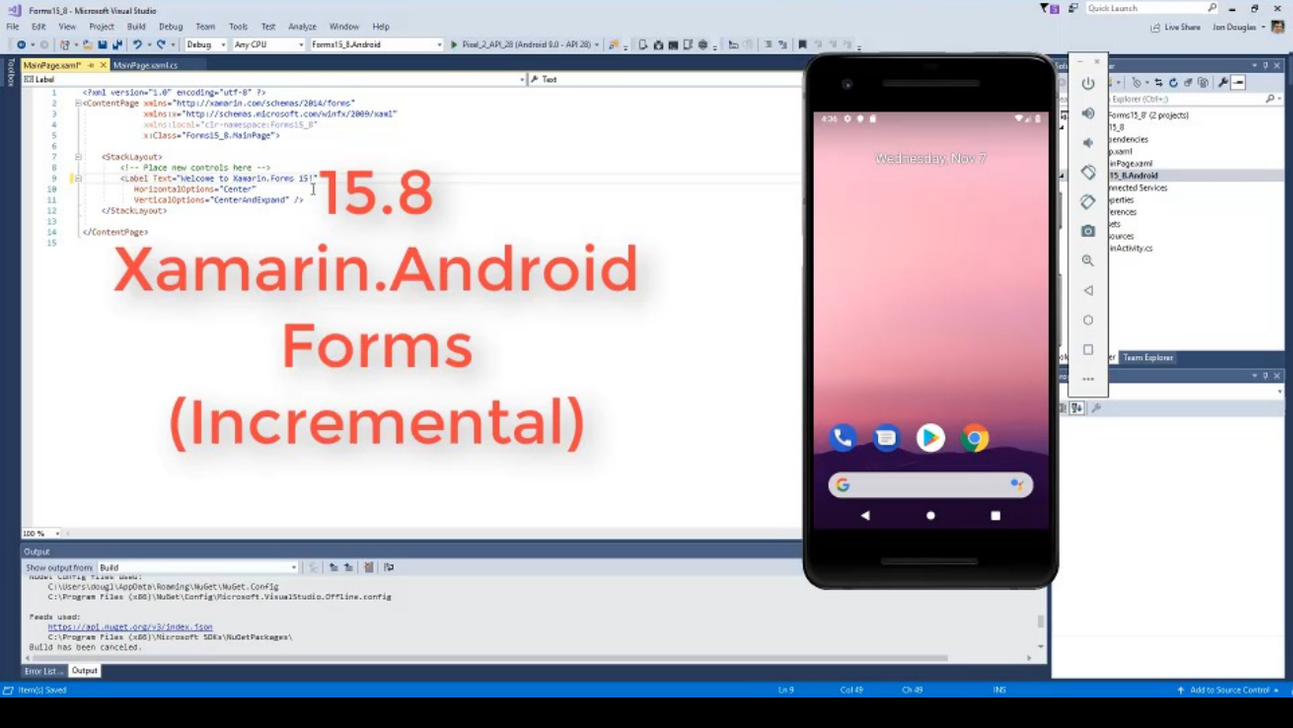
# Build Times

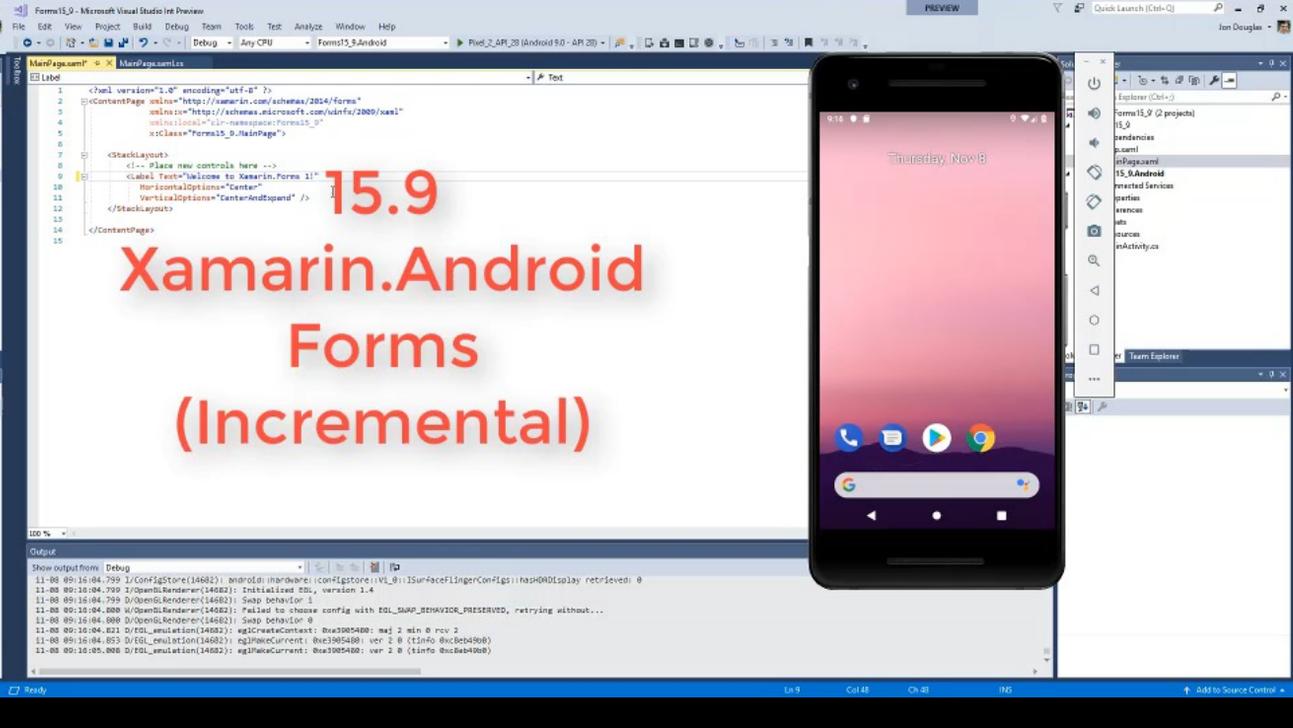
- First cold build
- Incremental build
- Deployment

https://github.com/xamarin/xamarin-android/wiki/Build-Performance-Results


15.8
Xamarin.Android
(Incremental)


15.9
Xamarin.Android
(Incremental)


15.8
Xamarin.Android
Forms
(Incremental)


15.9
Xamarin.Android
Forms
(Incremental)

# 13% talk about time waiting to view UI changes:

"Having to compile in order to preview"

"xaml previewer doesnt really work, lack of hot reload"

"Not able to see preview of the designed form"

"Live previewer NEVER works for me so I have to launch an emulator JUST to see my simple XAML changes."

Flexibility - Microsoft Visual Studio Preview

PREVIEW

Quick Launch

File Edit View Project Build Debug Team Tools Test Analyze Window Help

David Ortinau

Debug | Any CPU | Flexibility.Droid | pixel26 (Android 8.0 - API 26)

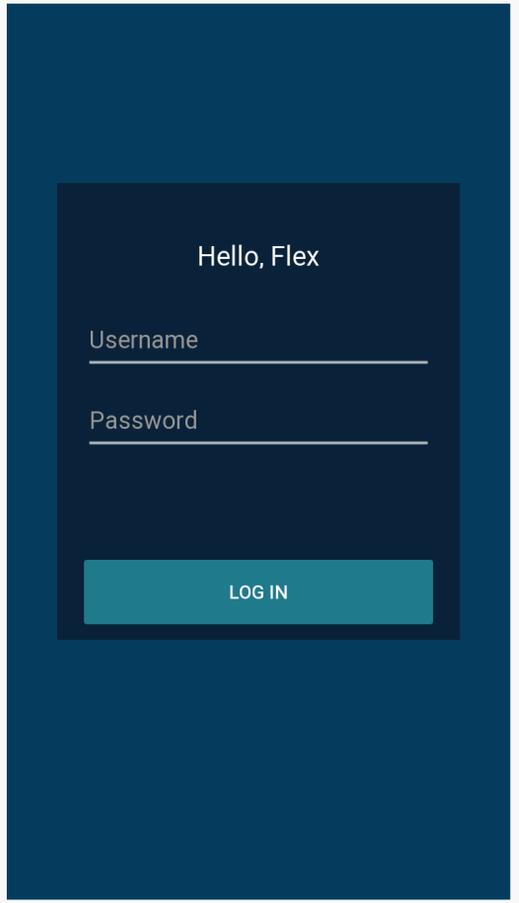
LoginPage.xaml | ShowPage.xaml

ContentPage | ContentPage

```
1   <?xml version="1.0" encoding="utf-8"?>
2   <ContentPage xmlns="http://xamarin.com/schemas/2014/forms"
3                xmlns:x="http://schemas.microsoft.com/winfx/2009/xaml"
4                x:Class="Flexibility.Shared.LoginPage"
5                xmlns:ios="clr-namespace:Xamarin.Forms.PlatformConfiguration
6                ios:Page.LargeTitleDisplay="Never">
7
8       <ContentPage.Resources>
9           <ResourceDictionary Source="Resources/FontRD.xaml"/>
10          <ResourceDictionary Source="Resources/EntryValidationRD.xaml"/>
11
12          <StyleSheet Source="Styles/LoginPage.css"/>
13      </ContentPage.Resources>
14
15      <FlexLayout StyleClass="body" x:Name="Form">
16          <FlexLayout x:Name="LoginForm" StyleClass="container">
17              <Label Text="Hello, Flex" StyleClass="logo" />
18              <FlexLayout StyleClass="login-form">
19                  <Entry x:Name="UserNameEntry"
20                         IsSpellCheckEnabled="false"
21                         Placeholder="Username"
22
23                         MaxLength="7" >
24                      <Entry.TextColor>
25                          <OnPlatform x:TypeArguments="Color">
26                              <On Platform="Android" Value="#f1f1f1"/>
27                          </OnPlatform>
28                      </Entry.TextColor>
```

177 % | David Ortinau, 176 days ago | 1 author, 9 changes

Device: Phone Tablet Platform: Android iOS




Error List Output

Ready | 0 | 15 | flexibility | master

Gorilla › iPhone 8 iOS 12.0 | Generated file successfully. 10 | Press '⌘.' to search

Solution

- Grial.Templates (master)
  - Grial.Templates
    - Getting Started
    - Dependencies
    - Controls
    - Flows
      - Ecommerce
        - ViewModels
        - Views
          - Templates
          - CheckoutPage.xaml
          - EcommerceMainPage.xaml
          - OrderConfirmationPage.xaml
          - ProductDetailPage.xaml
        - EcommerceNavigationPage.xaml
      - Movies
        - ViewModels
        - Views
          - Templates
          - FeaturedMoviesPage.xaml
          - MovieDetailPage.xaml (Gorilla)
          - MoviesMainPage.xaml
        - MoviesNavigationPage.xaml
      - Root
      - Tasks
        - ViewModels
        - Views
          - Templates
          - EmployeePerformanceDashboardPage.xam
          - EmployeeProfileDashboardPage.xaml
          - PerformanceDashboardMainPage.xaml
        - PerformanceDashboardNavigationPage.xaml
    - Helpers
    - Properties
    - Resources

MovieDetailPage.xaml

Preview

```
</ContentPage.Resources>

<ContentPage.Content>
    <Grid>
        <Grid x:Name="PictureHeader">
            <Grid.RowDefinitions>
                <RowDefinition Height="360" />
            </Grid.RowDefinitions>

            <ffimageloading:CachedImage
                Grid.Row="0"
                FadeAnimationEnabled="true"
                Source="{ Binding Source }"
                Aspect="AspectFill"
                DownsampleHeight="360"
                Opacity="1"
            />

            <!-- MOVIE BLUR BACKGROUND -->
            <ffimageloading:CachedImage
                Grid.Row="0"
                FadeAnimationEnabled="true"
                Source="{ Binding Source }"
                Aspect="AspectFill"
                HeightRequest="100"
                DownsampleHeight="20"
                Opacity="1"
                ScaleX="1.2"
            >
                <ffimageloading:CachedImage.Transformations>
                    <fftransformations:BlurredTransformation Radius="2"/>
                </ffimageloading:CachedImage.Transformations>

                <VisualElement.Behaviors>
                    <animations:AnimatedOpacity
                        Progress="{ Binding AbsoluteScrollY, Source={x:Reference scroll} }"
```

Content | Changes | Authors | Log | Merge

Test Results | Errors | Package Console | Application Output - Grial.Templates.iOS

```
2018-09-29 12:30:34.253239-0300 UXDivers.Grial.Templates.iOS[837:10765] SecTaskLoadEntitlements failed error=22
2018-09-29 12:30:34.253424-0300 UXDivers.Grial.Templates.iOS[837:10765] SecTaskCopyDebugDescription: UXDivers.Gr
2018-09-29 12:30:34.262008-0300 UXDivers.Grial.Templates.iOS[837:10765] libMobileGestalt MobileGestalt.c:890: MG
Thread started: <Thread Pool> #2
Thread started: <Thread Pool> #3
Thread started: <Thread Pool> #4
Thread started: <Thread Pool> #5
```

Tasks | Tool Output

iPhone 8 - 12.0

Carrier 12:35 PM




Play Movie

## Inception

A thief, who steals corporate secrets through use of dream-sharing technology, is given the inverse task of planting an idea into the mind of a CEO."

**Audience rating**

ACTION | THRILLER | SCIENCE FICTION | MYSTERY | ADVENTURE

**CAST:** Leonardo DiCaprio, Joseph Gordon-Levitt,

# Design and Runtime Iteration

- XAML Previewer improvements in VS2019
- Xamarin.Forms Property Editor in VS2019
- Interviews with developers about hot reload

- Live Reload (End of Preview) is still available (Win only)
- Live XAML www.livexaml.com ($)
- Gorilla Player http://gorillaplayer.com (FREE)
- XAMLator https://github.com/ylatuya/XAMLator (Open Source)

# Missing controls and properties (8%)

“Missing functionality offered by native controls (e.g. text color of entry cell)”

“A lot of missing features in shared libraries, therefore a lot of platform specific code required”

# Xamarin Essentials

A kit of essential cross platform APIs for your mobile apps – Native Android, Native iOS or Forms



Accelerometer

App Information

Battery State

Clipboard

Compass

Connectivity

Data Transfer

Device Information

File System Helpers

Email

Geocoding

Geolocation

Gyroscope

Magnetometer

Open Browser

Phone Dialer

Preferences

Screen lock

Secure Storage

SMS

Text to Speech

Version tracking

Vibrate

# F100: Addressing the Little Things

Project Board

SearchBar IsSpellCheckEnabled

ImeOptions

DesignModeEnabled

Android Bottom Tabs

WebView mixed content (Android)

WebView evaluate javascript

Binding Mode OneTime

ProgressBar colors

Switch colors

Slider colors

Hide scroll bars

Button AccessKey

TabbedPage Icons (UWP)

Drop Shadows (iOS/UWP)

App DI Container Hooks

Button Padding

Page TitleView

BoxView rounded corners

WKWebView for iOS

Entry/Editor

- IsSpellCheckEnabled
- MaxLength
- DetectReadingOrderFromContent (UWP)
- Text prediction options
- Autocapitalization
- AutoResizable Editor
- ButtonType for return button
- Position and Color of caret
- Placeholder color

Label

- Bindable Spans
- Commandable Spans, Span Gestures
- LineHeight

ListView

- Full width separators
- SelectionMode

**Coming Soon**

Label MaxLines

Entry Read-Only

ImageButton

GIF Support

Button and Label Parity

PinsSource on Forms.Map

WebView Zoom Controls (Android)

**Help Wanted**

Rounded Corners

Rounded Image

ProgressBar.Height

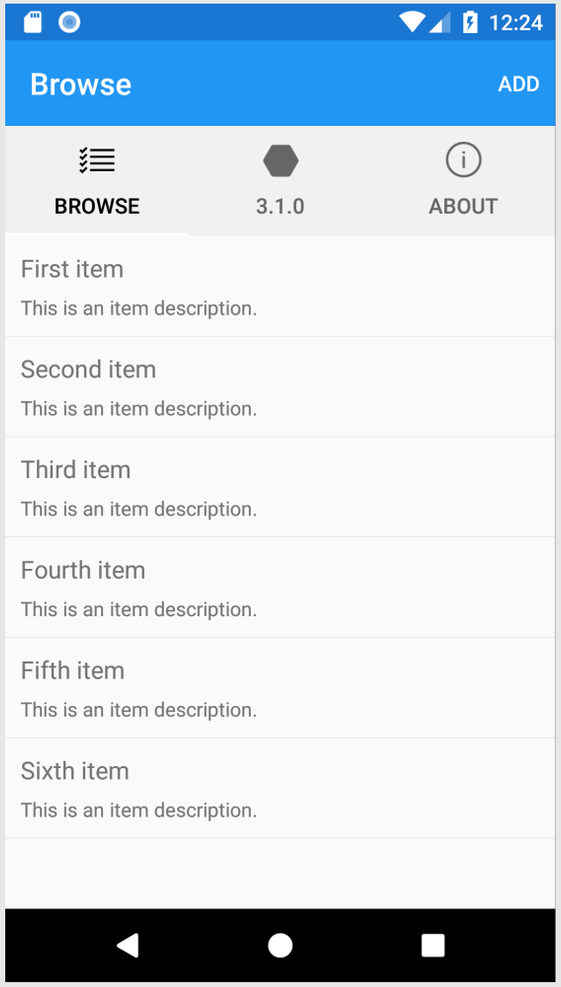
InkWell Ripple Effect

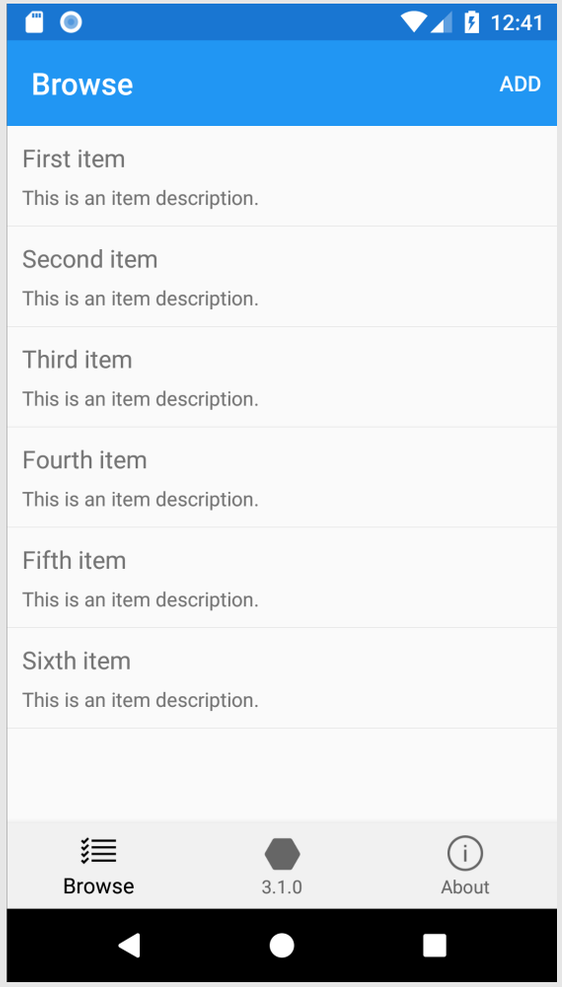
Size WebView to Content

Floating Action Button

Android Theme Capitalization override

# Android Bottom Tabs

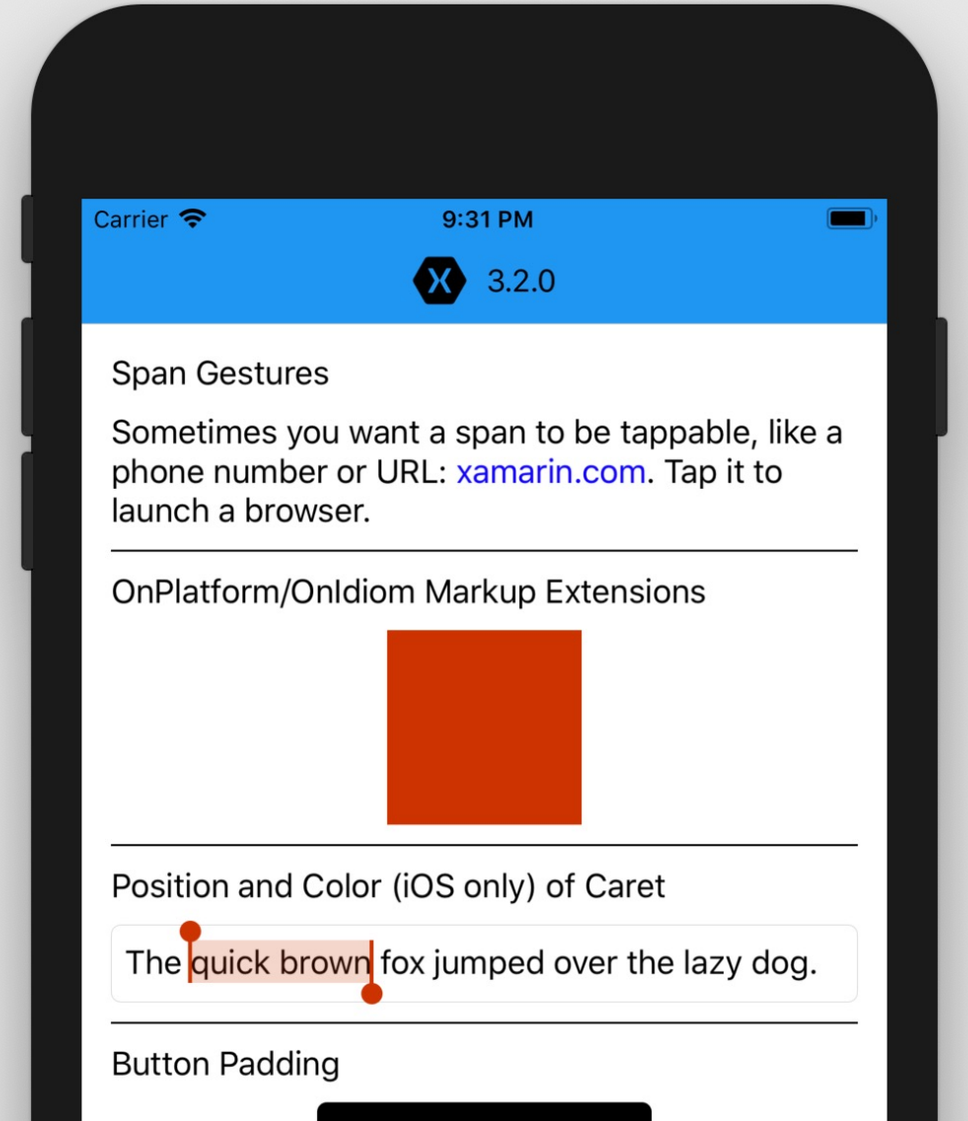



```
<TabbedPage …
            BarBackgroundColor="#F1F1F1"
            xmlns:android="clr-
namespace:Xamarin.Forms.PlatformConfiguration.AndroidSpecific;assembly=
Xamarin.Forms.Core"
            android:TabbedPage.ToolbarPlacement="Bottom"
            android:TabbedPage.BarItemColor="#666666"
            android:TabbedPage.BarSelectedItemColor="Black">

```

# TitleView



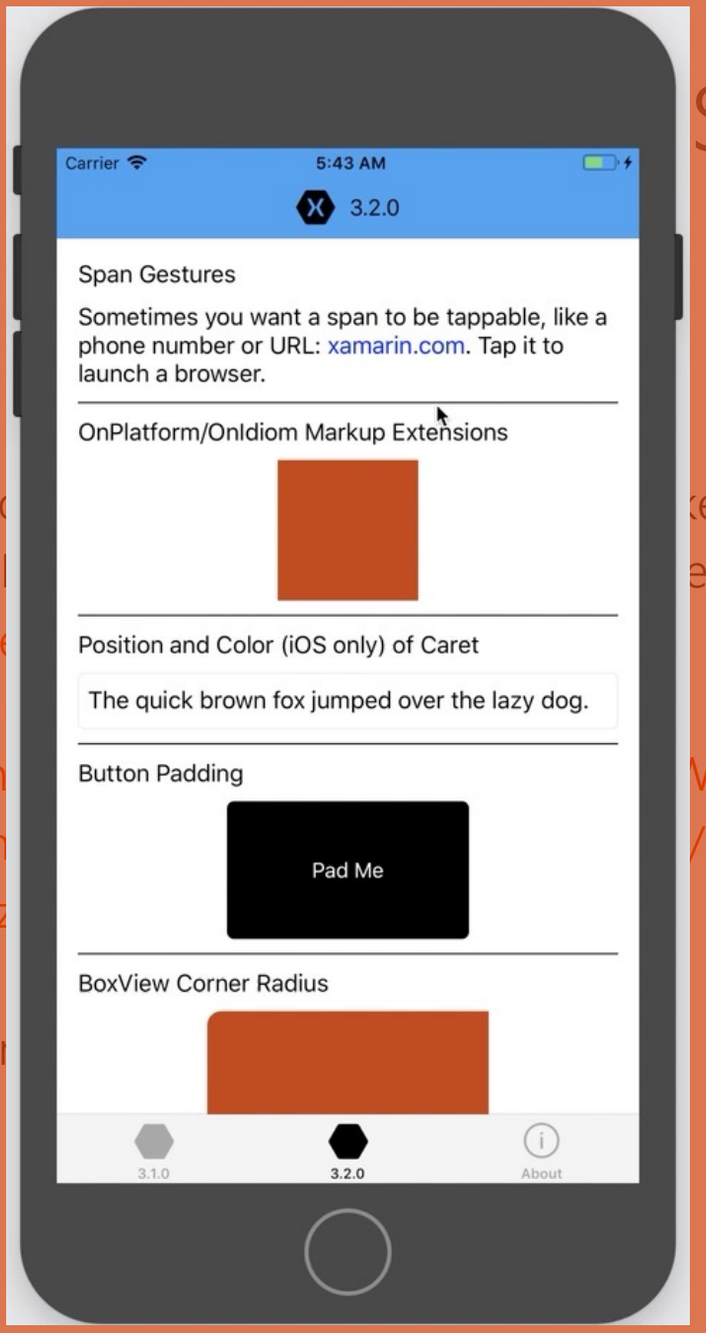

```
<ContentPage>
    <NavigationPage.TitleView>
        <StackLayout
            Orientation="Horizontal"
            VerticalOptions="Center" Spacing="10">
            <Image Source="iconXamagon.png">
                <Image.GestureRecognizers>
                    <TapGestureRecognizer
                        Tapped="Handle_Tapped"/>
                </Image.GestureRecognizers>
            </Image>
            <Label Text="3.2.0"
                FontSize="16"
                TextColor="Black"
                VerticalTextAlignment="Center" />
        </StackLayout>
    </NavigationPage.TitleView>
    ...
</ContentPage>
```

# Bindable Span

```
<Label>
    <Label.FormattedText>
        <FormattedString>
            <Span Text="Welcome " />
            <Span Text="{Binding YourName}" FontAttributes="Bold"/>
            <Span Text=" to the Xamarin.Forms 3.1.0 playground!" />
        </FormattedString>
    </Label.FormattedText>
</Label>
```

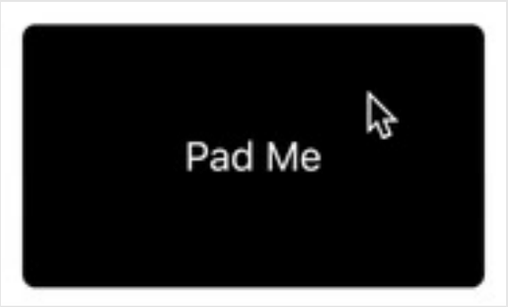
# Commandable

```
<Label>
    <Label.FormattedText>
      <FormattedString>
        <Span Text="Sometimes yo                                    ke a phone number or URL: " />
        <Span Text="{Binding Url, M                                 e">
          <Span.GestureRecognize
            <TapGestureRecog
              Command="{Bin                                         Way}"
              CommandParam                                          />
          </Span.GestureRecogniz
        </Span>
        <Span Text=". Tap it to laun
      </FormattedString>
    </Label.FormattedText>
</Label>
```



# Button Padding

```
<Button Text="Pad Me"
        Padding="60,40"
        BackgroundColor="Black"
        TextColor="White"/>
```



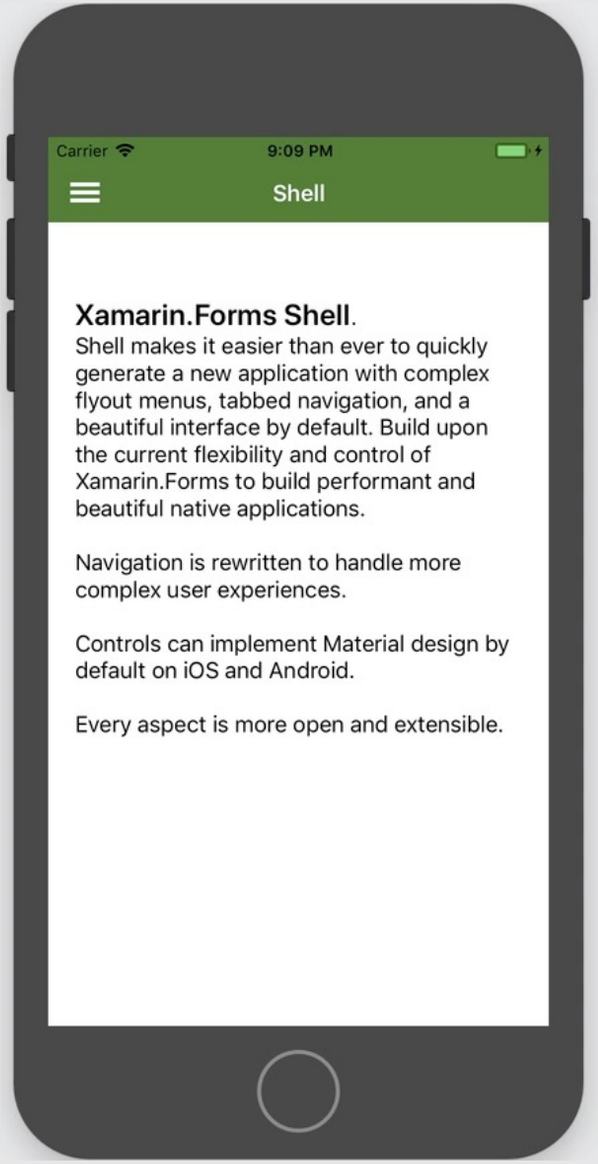
# OnPlatform & OnIdiom XAML Extensions

- Concise syntax
- Supports Default value
- Support Converter and ConverterParamter

```
<BoxView
    FlexLayout.AlignSelf="Center"
    WidthRequest="100"
    HeightRequest="100"
    BackgroundColor="{OnPlatform Android=#333333, iOS=#CC3300, Default=Green}" />
```

# Xamarin.Forms 4.0

# Xamarin.Forms Shell




- Everything you can already do today in Xamarin.Forms
- Performant by default
- Same design on iOS and Android via MaterialShell
- Platform design via Shell
- Easily describe your entire app in one file
- Flyout and Tabbed navigation
- URL routed navigation, deep linking
- Back navigation handling
- Integrated Search handler
- Snackbar
- Bottom Sheet
- Floating Action Button
- Left Bar Button
- Screen segues
- Screen transitions

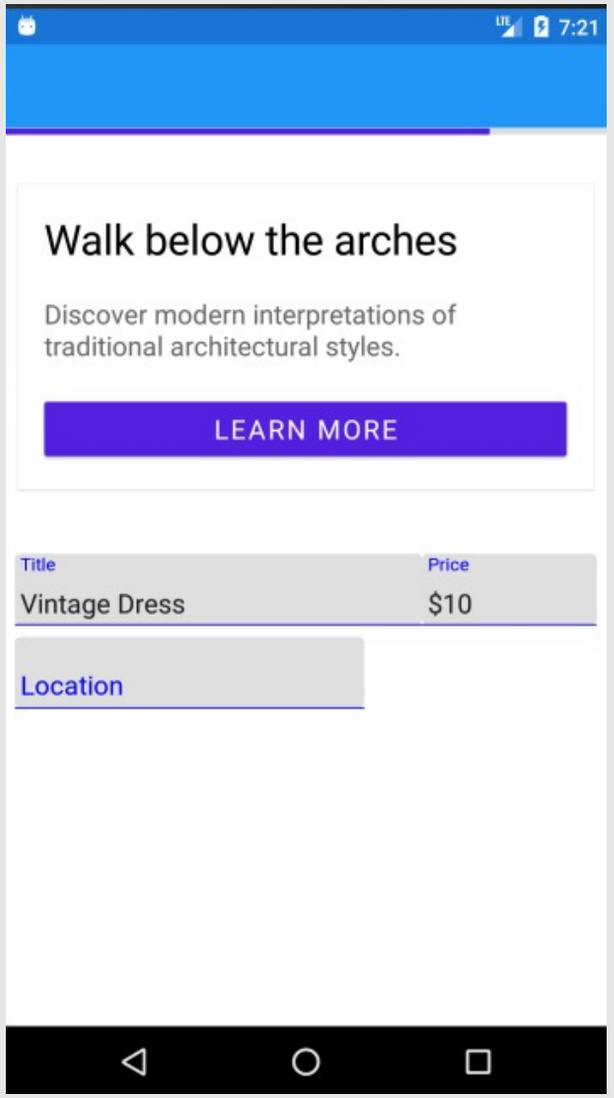
# Shell XAML Demo

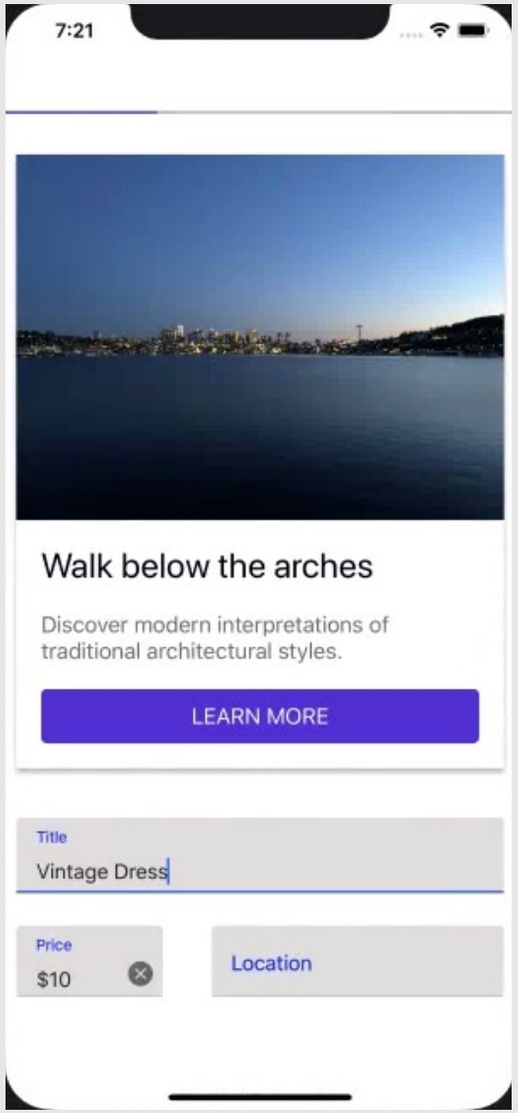
https://github.com/davidortinau/Gastropods/

# Visual

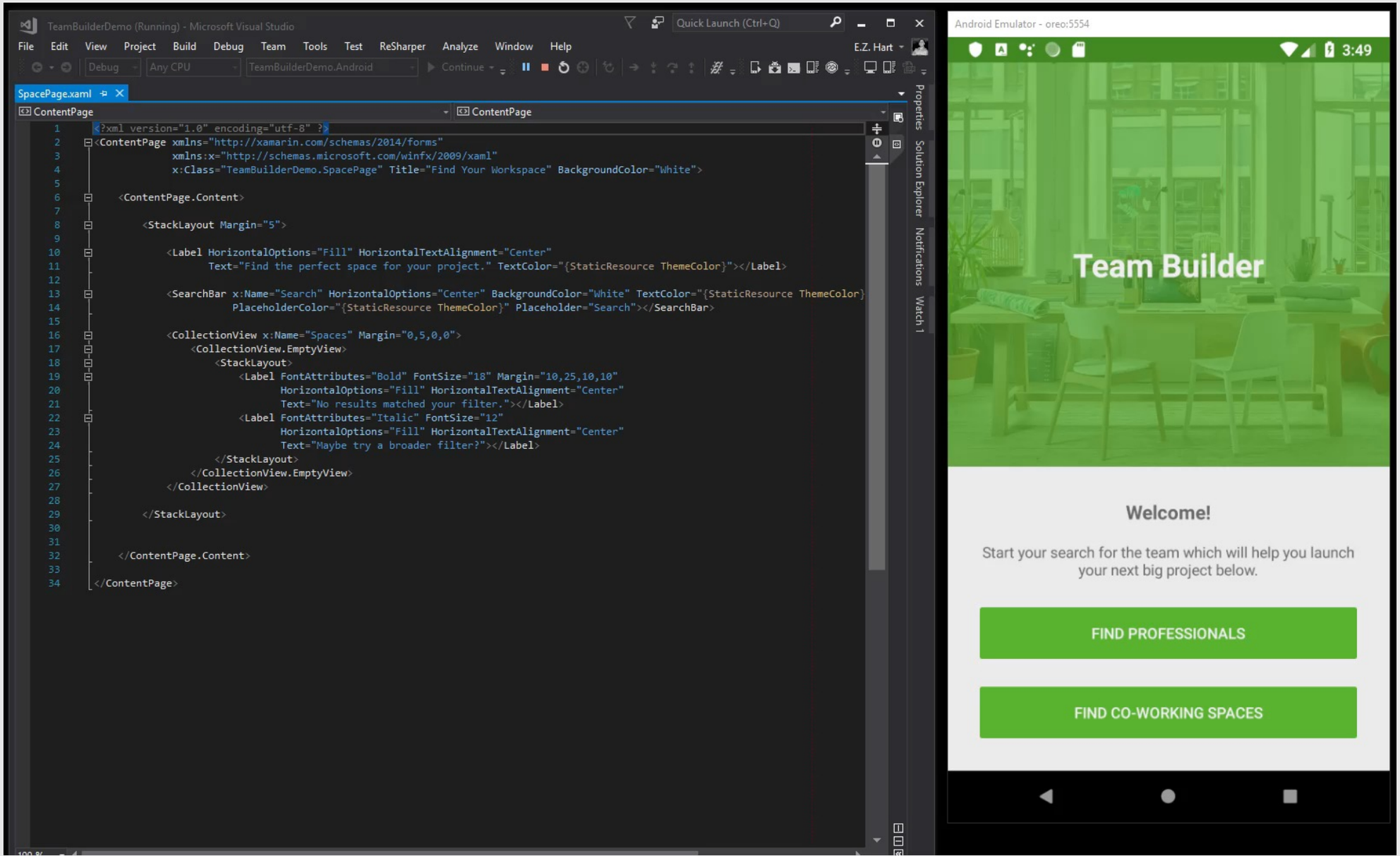
```
<ContentPage
  Visual="Material“
..>

<Entry
  Visual="Material“
..>
```



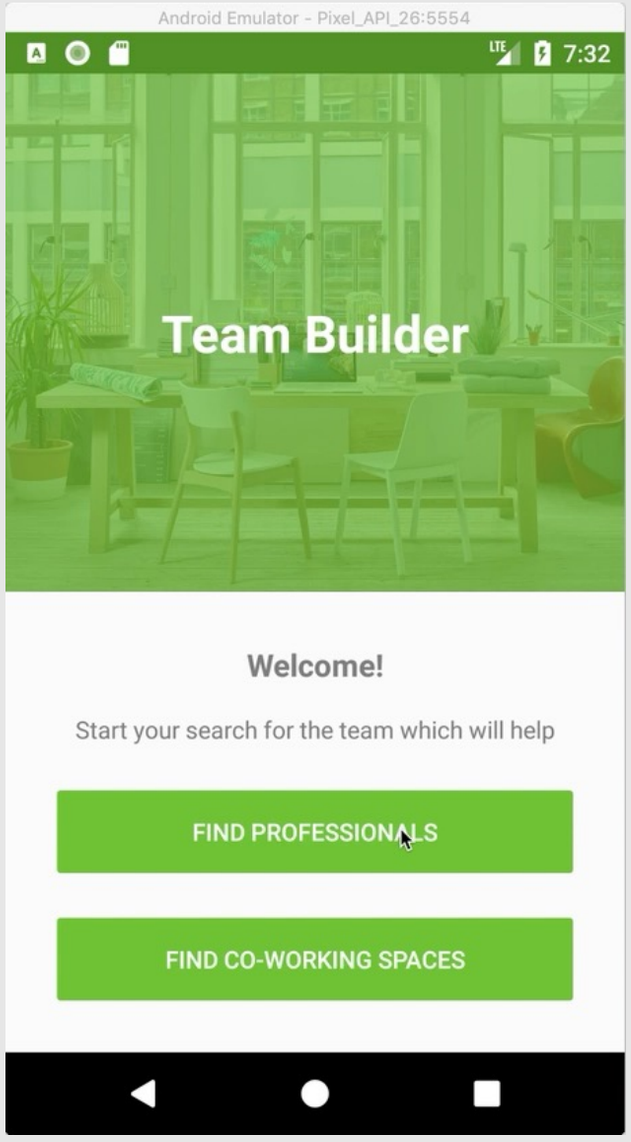
# CollectionView



# XAML

```
<CollectionView x:Name="Spaces" Margin="0,5,0,0">
    <CollectionView.ItemsLayout>
        <GridItemsLayout Span="2" Orientation="Vertical"/>
    </CollectionView.ItemsLayout>
    <CollectionView.EmptyView>
        <StackLayout>
            <Label FontAttributes="Bold" FontSize="18" Margin="10,25,10,10"
                HorizontalOptions="Fill" HorizontalTextAlignment="Center"
                Text="No results matched your filter."></Label>
            <Label FontAttributes="Italic" FontSize="12"
                HorizontalOptions="Fill" HorizontalTextAlignment="Center"
                Text="Maybe try a broader filter?"></Label>
        </StackLayout>
    </CollectionView.EmptyView>
</CollectionView>
```

# CarouselView



# XAML

```
<CarouselView x:Name="CarouselView">
    <CarouselView.ItemTemplate>
        <DataTemplate>
            <local:Card />
        </DataTemplate>
    </CarouselView.ItemTemplate>
</CarouselView>
```



# Connect() 2018

December 4
https://www.microsoft.com/en-us/connectevent/

Microsoft

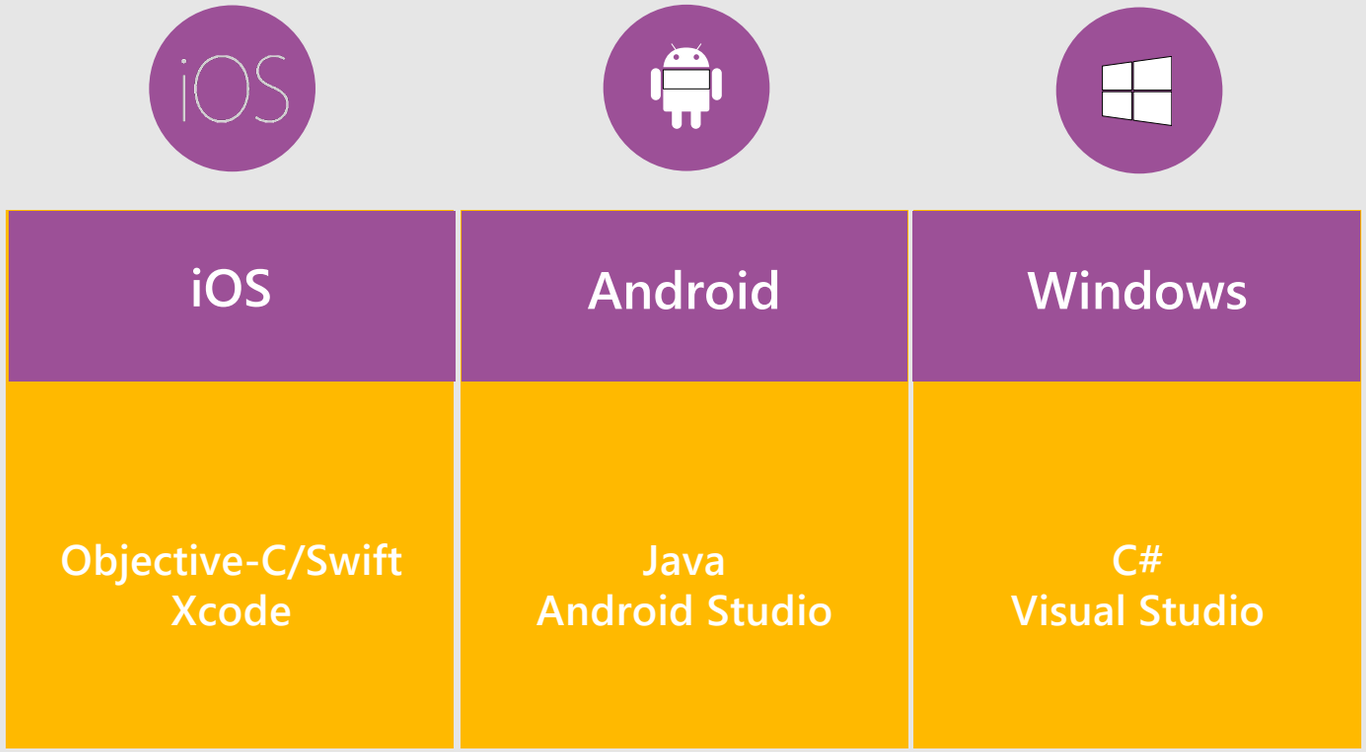
© Copyright Microsoft Corporation. All rights reserved

# What is Native?



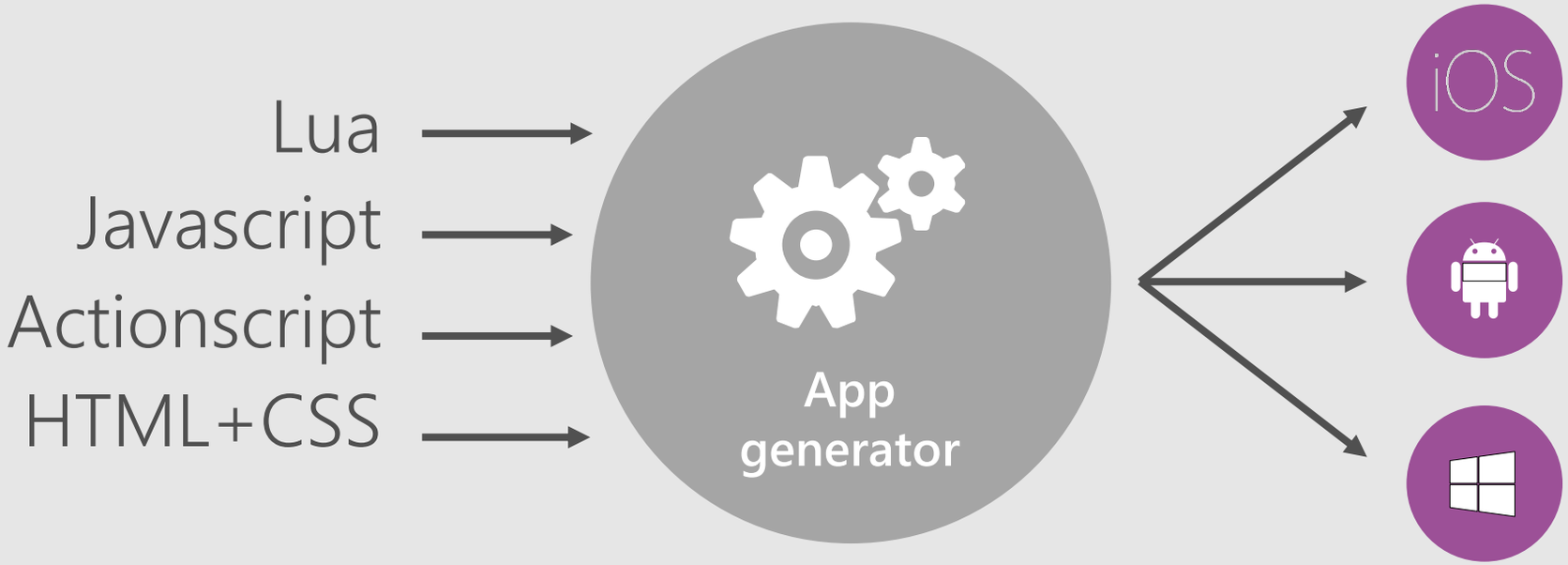
# Architecting Mobile Apps

# Silo approach



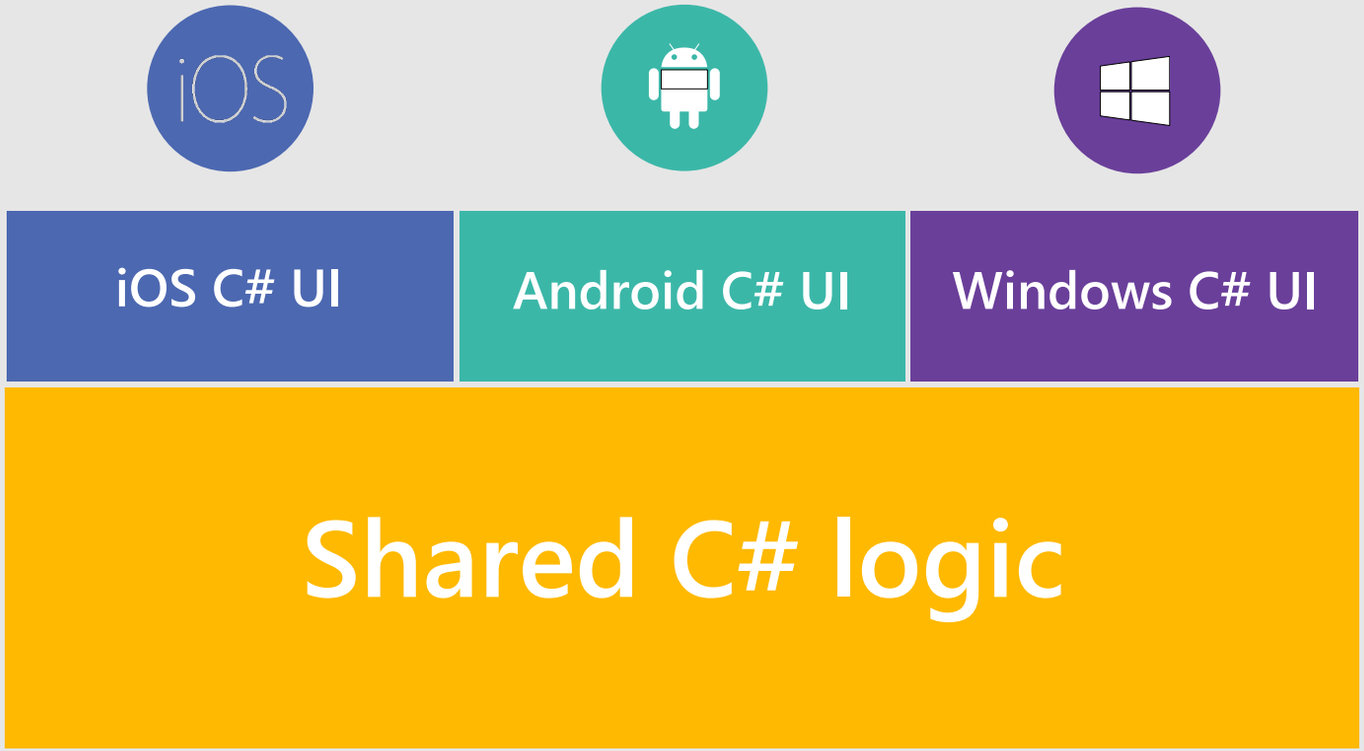

No shared code • Many languages and development environments • Multiple teams

# Write once, run anywhere




**Limited native API access • Slow performance • Poor user experience**

# Xamarin's unique approach




Shared C# codebase • 100% native API access • High performance

# User Interface

# Xamarin Native approach

iOS C# | Android C# | Windows C#

Shared C# Logic

3 Native User Interfaces

Shared App Logic



# Xamarin.Forms - Cross Platform UI





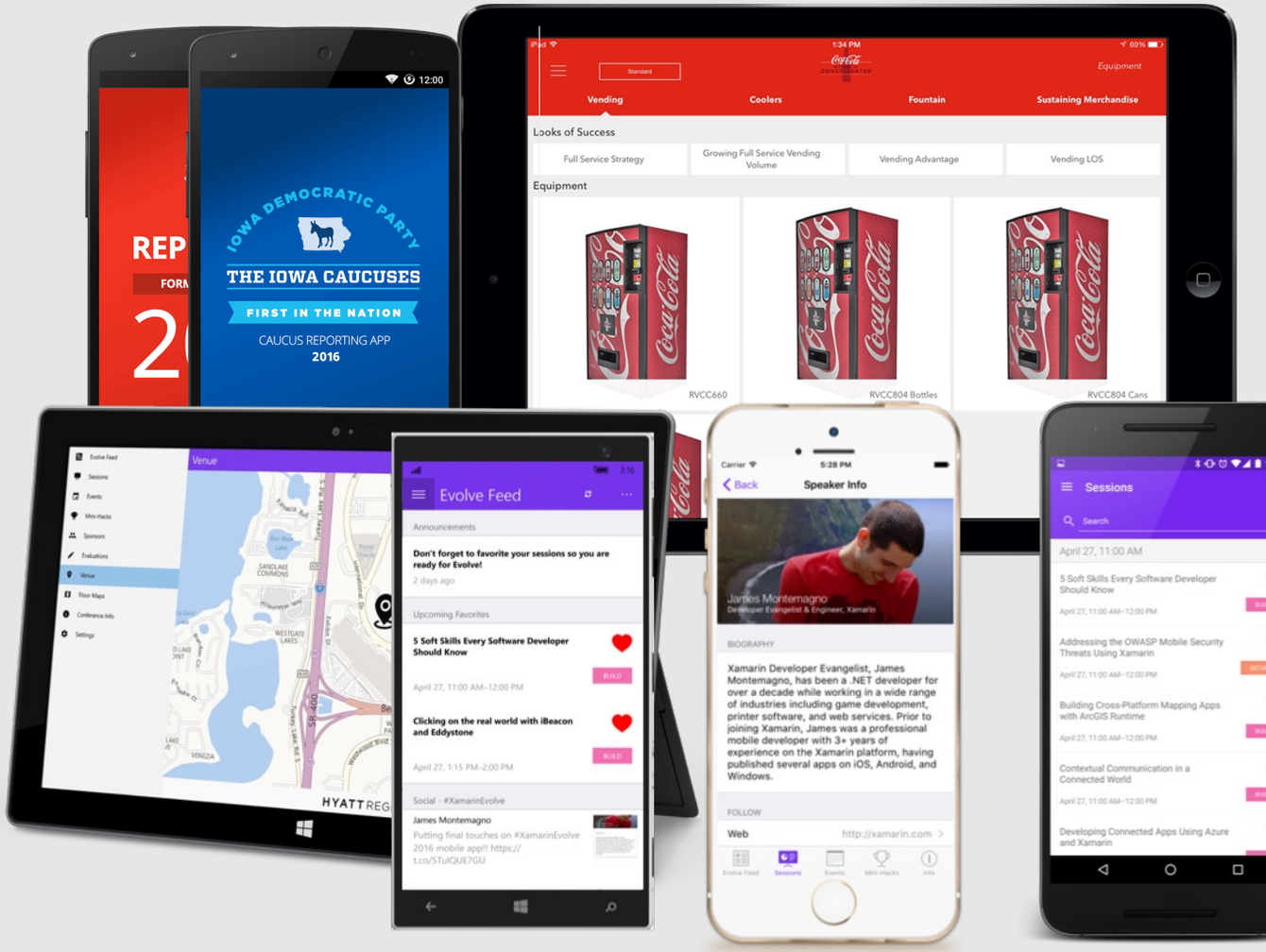



## Shared UI Code

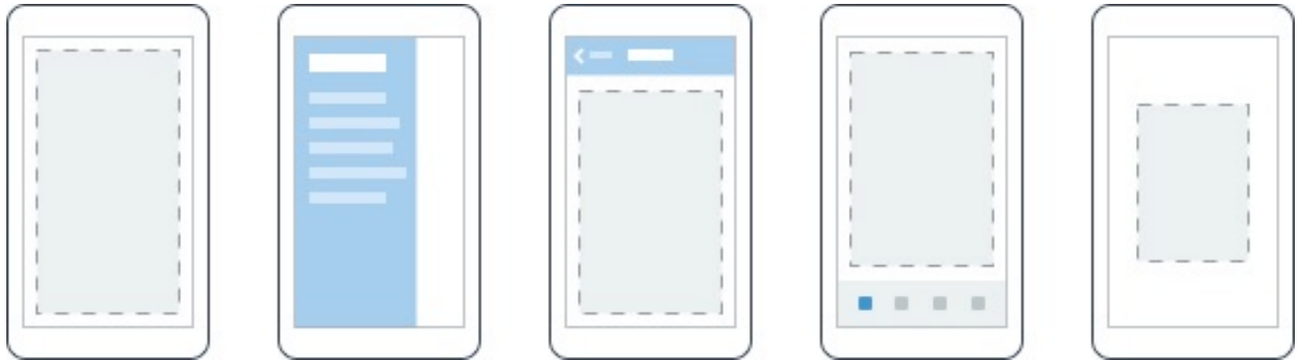
Shared C# Logic

Shared Native User Interface

Shared App Logic

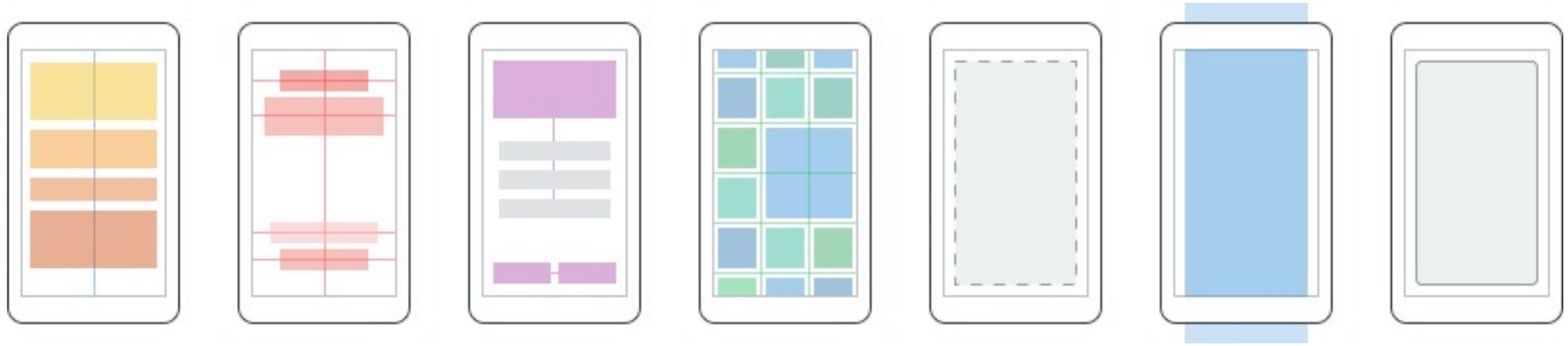


# Pages



Content MasterDetail Navigation Tabbed Carousel

# Layouts



Stack Absolute Relative Grid ContentView ScrollView Frame

# Controls

| | | | | |
|---|---|---|---|---|
| ActivityIndicator | BoxView | Button | DatePicker | Editor |
| Entry | Image | Label | ListView | Map |
| OpenGLView | Picker | ProgressBar | SearchBar | Slider |
| Stepper | TableView | TimePicker | WebView | EntryCell |
| ImageCell | SwitchCell | TextCell | ViewCell | |

# Native UI

## Forms elements are realized via platform renderers.

| | ContentPage | Label | Picker |
|---|---|---|---|
| Android | Activity / Fragment | TextView | AlertDialog + LinearLayout + EditText + NumberPicker |
| iOS | UIViewController | UILabel | UITextField + UIPickerView + UIToolbar + UIBarButtonItem |
| UWP | Page | TextBlock | ComboBox |

developer.xamarin.com/guides/xamarin-forms/application-fundamentals/custom-renderer/renderers/

# Bundled Services

DependencyService

MessagingCenter

Navigation

Themes

# 3rd Party App Frameworks

MvvmCross

MvvmLight

Prism

FreshMvvm

ReactiveUI

And more…

# Xamarin.Forms Ecosystem


















# Progressive Enhancement

# Progressive enhancement

## From standard experiences to total control



# Styling a Forms Element

```
<Style x:Key="EntryStyle" TargetType="{x:Type Entry}">
    <Setter Property="TextColor" Value="{StaticResource WhiteColor}" />
    <Setter Property="PlaceholderColor" Value="{StaticResource WhiteColor}" />
    <Setter Property="FontSize" Value="{StaticResource LargeSize}" />
    <Setter Property="FontFamily" Value="{StaticResource KarlaRegular}" />
    <Setter Property="HorizontalOptions" Value="FillAndExpand" />
    <Setter Property="FontAttributes" Value="None" />
    <Setter Property="Opacity" Value="0.85" />
    <Setter Property="effects:LineColorEffect.ApplyLineColor" Value="True" />
    <Style.Triggers>
        <Trigger TargetType="Entry"
            Property="IsFocused"
            Value="True">
            <Setter Property="Opacity" Value="1" />
        </Trigger>
    </Style.Triggers>
</Style>
```

# Styling a Forms Element

```
<Entry
      Text="{Binding Email.Value}"
      Placeholder="Email"
      Style="{OnPlatform Windows=UwpEntryStyle, Default=EntryStyle} " />
```

# Styling a Forms Element

```
.entry {
    color: #FFF;
    xf-placeholder-color: #FFF;
    font-size: 18px;
    font-family: KarlaRegular;
    xf-horizontal-options: FillAndExpand;
    opacity: 0.85;
}

.uwpEntry {…}

<Entry
        Text="{Binding Email.Value}"
        Placeholder="Email"
       class= "{OnPlatform Windows=uwpEntry, Default=entry}" />
```

# Platform-Specific

```
<ContentPage
        ...
        xmlns:ios="clr-namespace:Xamarin.Forms.PlatformConfiguration.iOSSpecific;
                   assembly=Xamarin.Forms.Core”>

        <BoxView x:Name="boxView"
                 ios:VisualElement.BlurEffect="ExtraLight"
                 HeightRequest="300"
                 WidthRequest="300" />

</ContentPage>
```

# Effects - XAML

```
<Label Text="Label Shadow Effect">
    <Label.Effects>
        <local:ShadowEffect Radius="5" DistanceX="5" DistanceY="5">
            <local:ShadowEffect.Color>
                <OnPlatform x:TypeArguments="Color"
                    iOS="Black"
                    Android="White"
                    WinPhone="Red" />
            </local:ShadowEffect.Color>
        </local:ShadowEffect>
    </Label.Effects>
</Label>
```

# Effects - RoutingEffect

```
public class ShadowEffect : RoutingEffect
{
	public float Radius { get; set; }

	public Color Color { get; set; }

	public float DistanceX { get; set; }

	public float DistanceY { get; set; }

	public ShadowEffect () : base ("MyCompany.LabelShadowEffect")
	{
	}
}
```

# Effects - PlatformEffect

```
[assembly:ResolutionGroupName ("MyCompany")]
[assembly:ExportEffect (typeof(LabelShadowEffect), "LabelShadowEffect")]
namespace EffectsDemo.iOS
{
    public class LabelShadowEffect : PlatformEffect
    {
        protected override void OnAttached ()
        {
            try {
                var effect = (ShadowEffect)Element.Effects.FirstOrDefault (e => e is ShadowEffect);
                if (effect != null) {
                        Control.Layer.CornerRadius = effect.Radius;
                        Control.Layer.ShadowColor = effect.Color.ToCGColor ();
                        Control.Layer.ShadowOffset = new CGSize (effect.DistanceX, effect.DistanceY);
                        Control.Layer.ShadowOpacity = 1.0f;
                }
            } catch (Exception ex) {
                Console.WriteLine ("Cannot set property on attached control. Error: ", ex.Message);
            }
        }
}
```

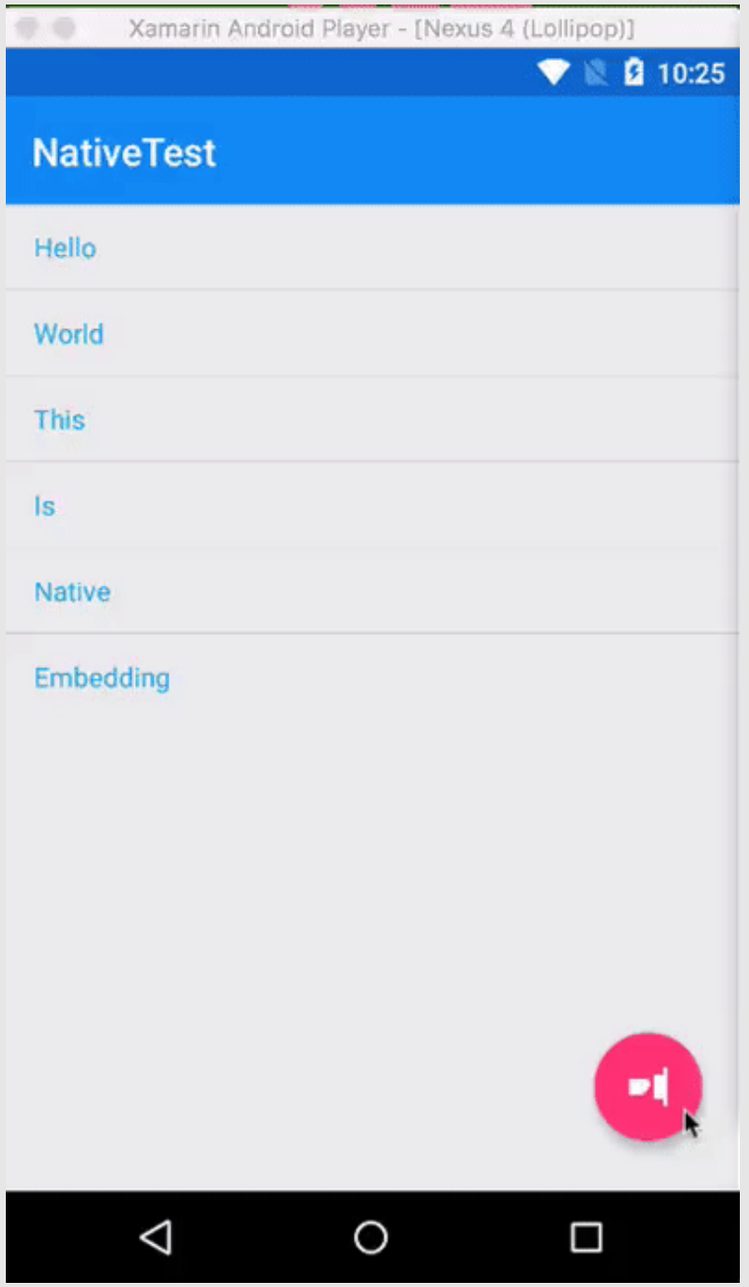
# Demo

Progressive Enhancement

# Native Control Embedding

# Floating Action Button

An Android-only control that has no Xamarin.Forms abstraction.



# #if to target only the desired platform

```
#if __ANDROID__
using Xamarin.Forms.Platform.Android;
using NativeTest.Droid; //Your Namespace
using Android.Views;
#endif
```

# Add the FAB

```
#if __ANDROID__
var fab = new CheckableFab(Forms.Context){  UseCompatPadding = true};

fab.SetImageResource(Droid.Resource.Drawable.ic_fancy_fab_icon);

fab.Click += async (sender, e) =>{
      await Task.Delay(3000);
      await MainPage.DisplayAlert("Native FAB Clicked", "Whoa!", "OK");
};
stack.Children.Add(fab);

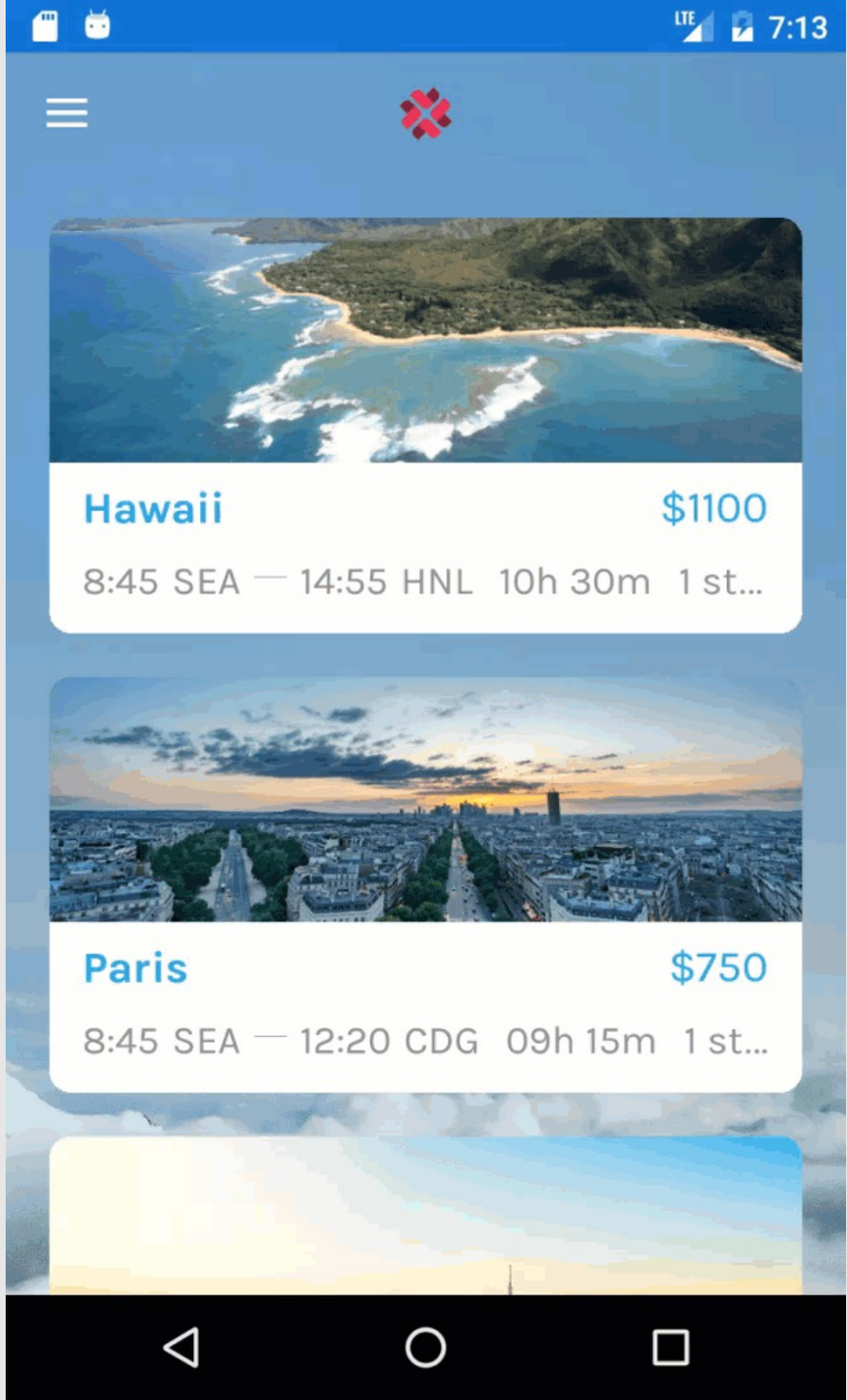
absolute.Children.Add(stack); // Overlay the FAB in the bottom-right
... // set layout and add to UI
#endif
```

# Gestures

# Gestures

Xamarin.Forms provides easy-to-use gestures.

- TapGestureRecognizer
- PinchGestureRecognizer
- PanGestureRecognizer



# TapGestureRecognizer

```
var tapGestureRecognizer = new TapGestureRecognizer();
GestureRecognizers.Add(tapGestureRecognizer);
tapGestureRecognizer.Tapped += AnimateCardClick;

private async void AnimateCardClick(object sender, EventArgs e)
{
      var contentView = (ContentView)sender;
      var item = (Grid)contentView.Content;
      await item.ScaleTo(0.95, 50, Easing.CubicOut);
      await item.ScaleTo(1, 50, Easing.CubicIn);
}
```

# What's Next for Xamarin.Forms?

# 2.4.0

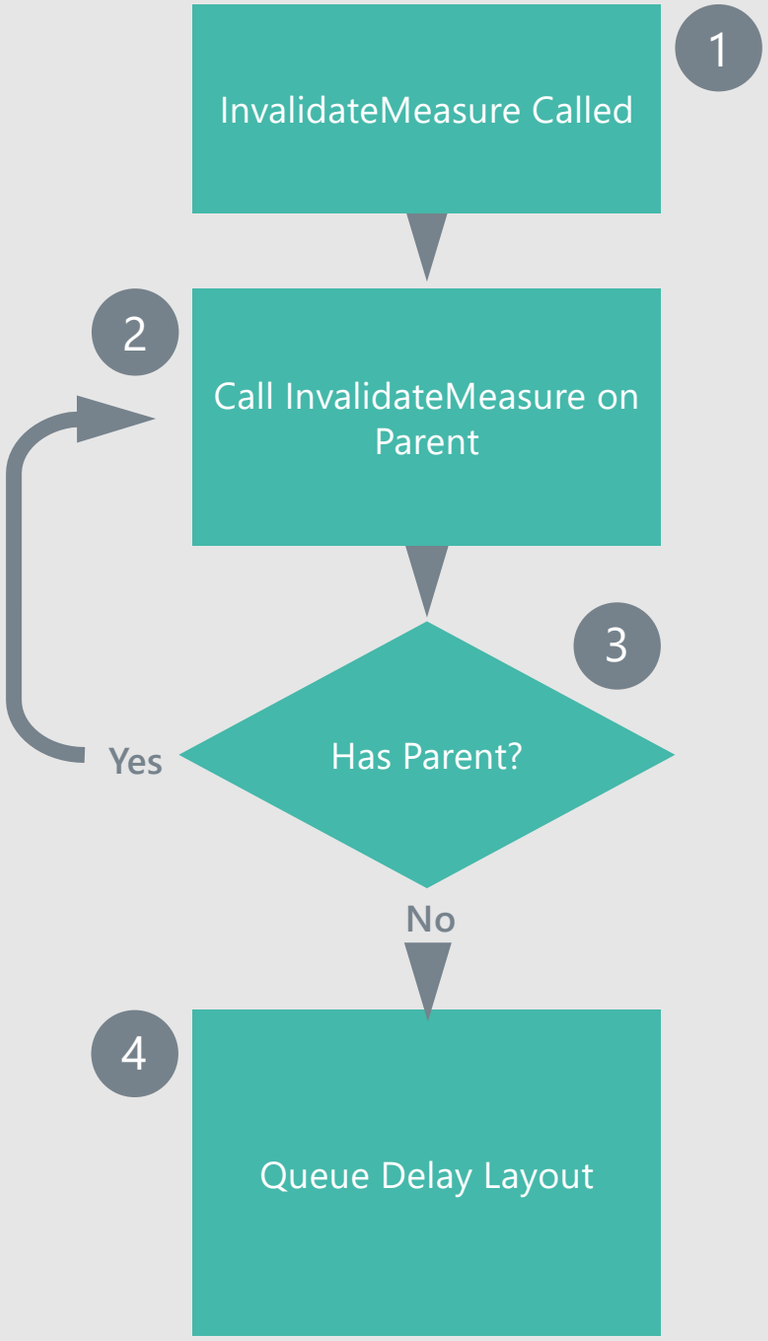
- .NET Standard 2.0
- Stability and Performance
- Over 150 fixes
- New Feature Flags API
- Android Fast Renderers: Button, Image, Label
- macOS Preview

# vNext

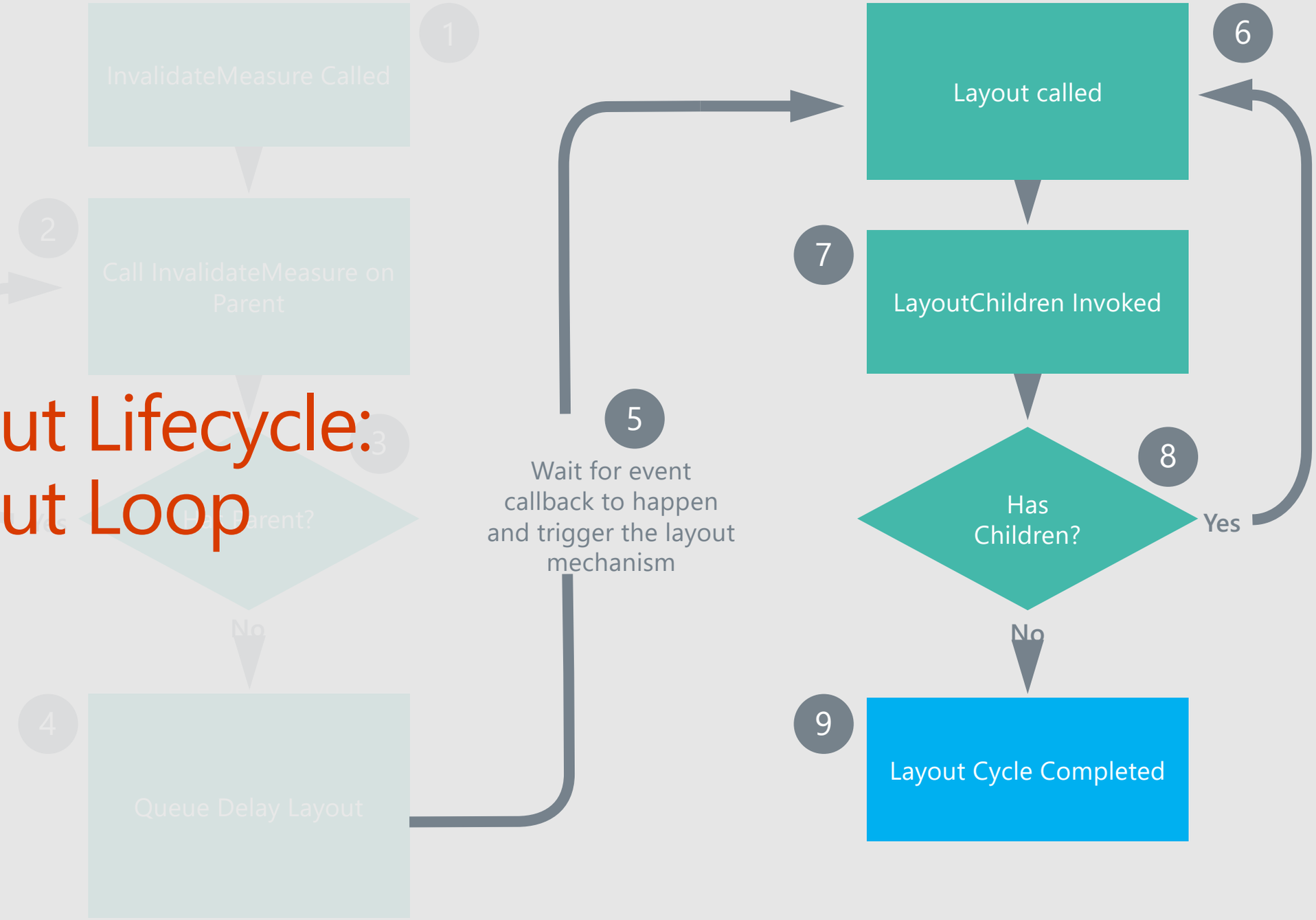
- Page Embedding
- Layout Compression
- Fast Renderers
- Right-to-Left
- ListView enhancements
- Gesture enhancements
- FlexLayout
- macOS, GTK

# Layout Compression

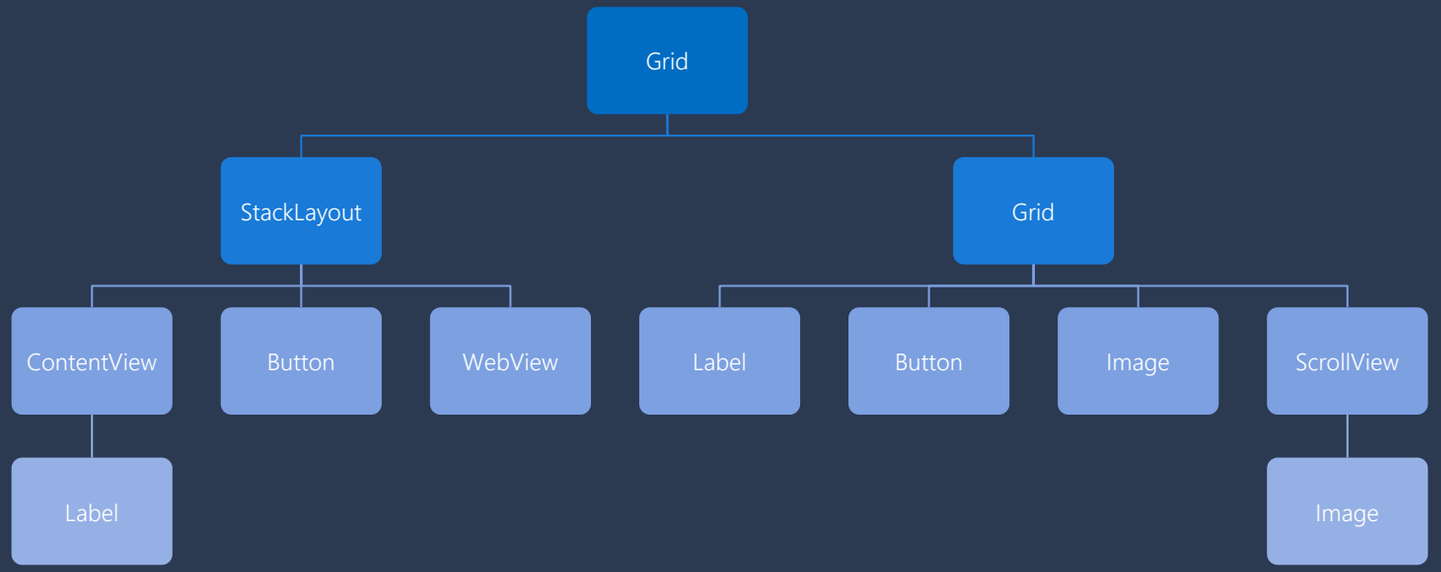
# Layout Lifecycle: Invalidation Loop



# Layout Lifecycle: Layout Loop

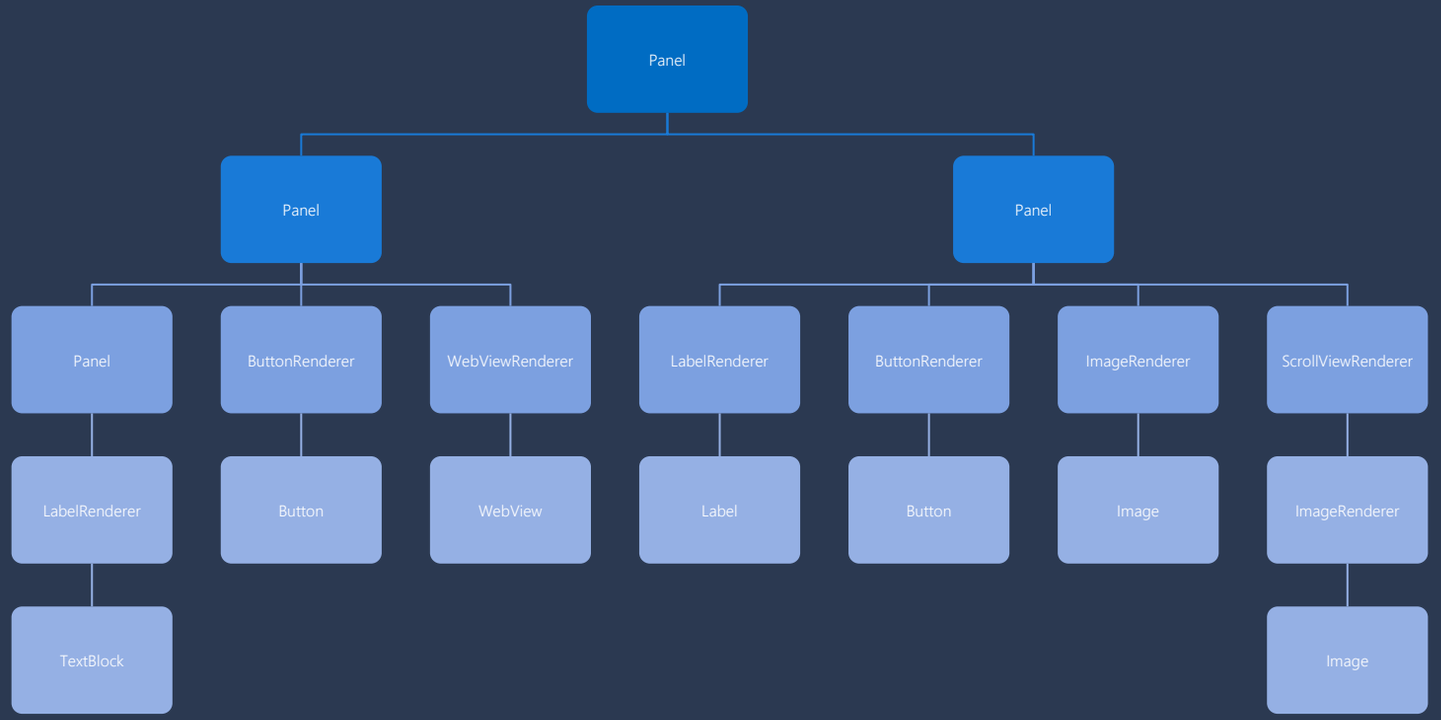


# Forms




12

# Platform



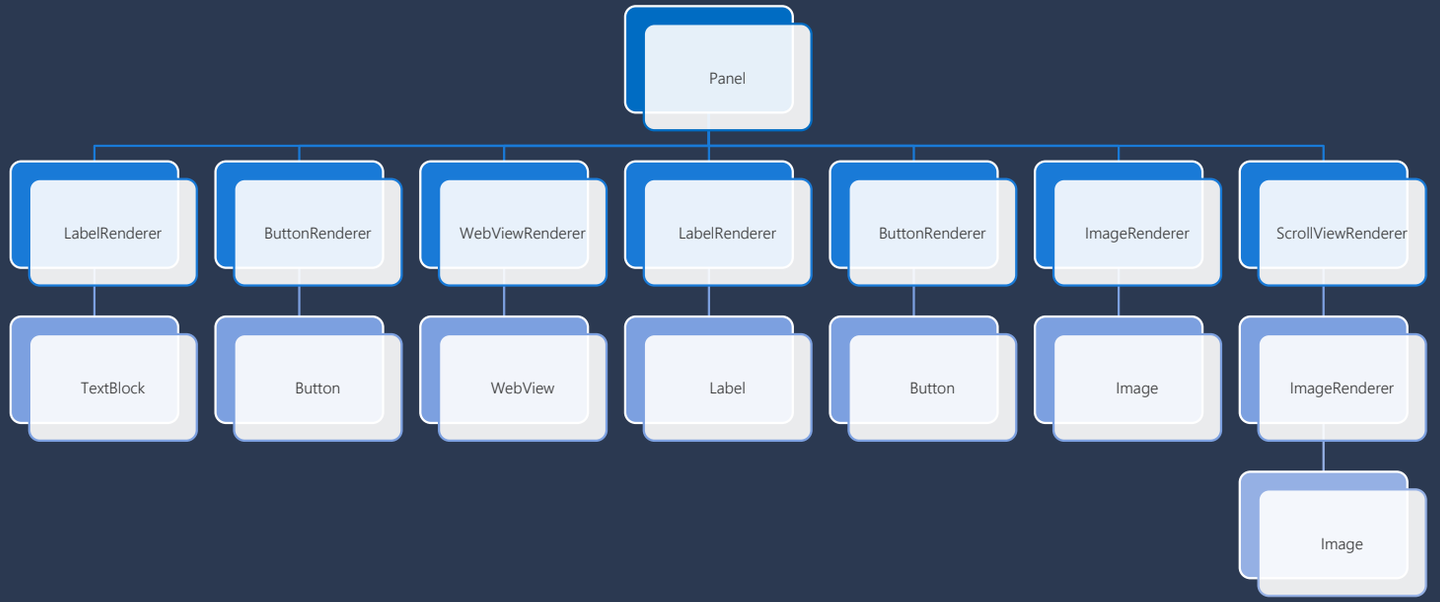

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# Platform+Compression

12

19

16



# Fast Renderers

# Forms

Button

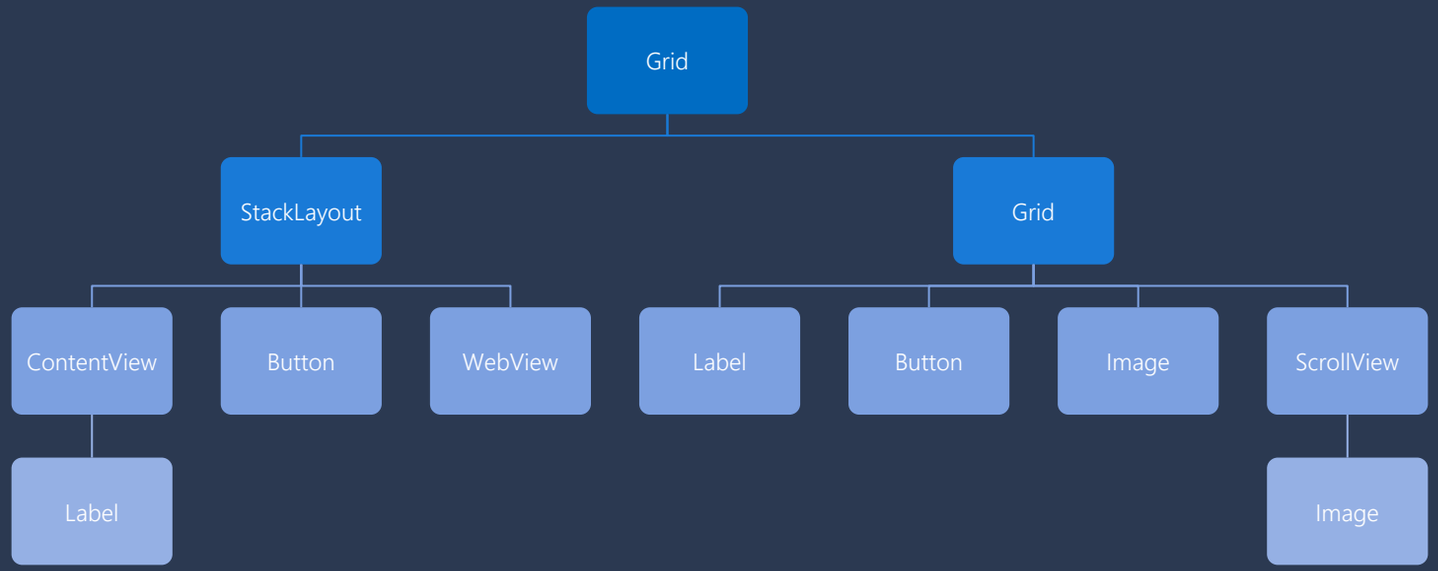
---

# Platform

ButtonRenderer

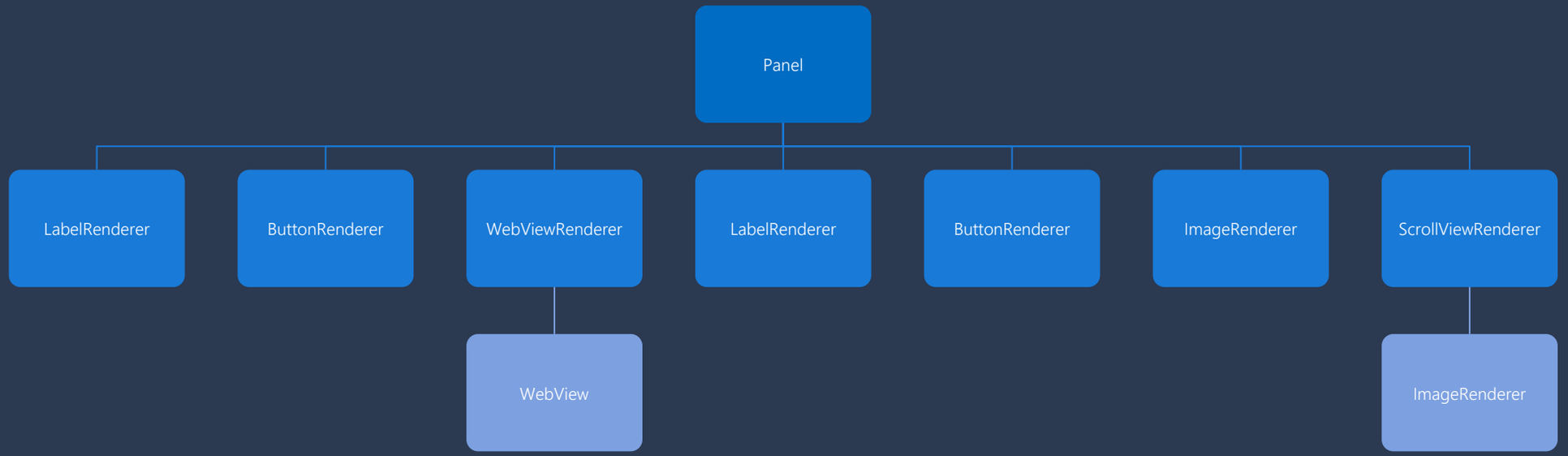
Button

# Forms



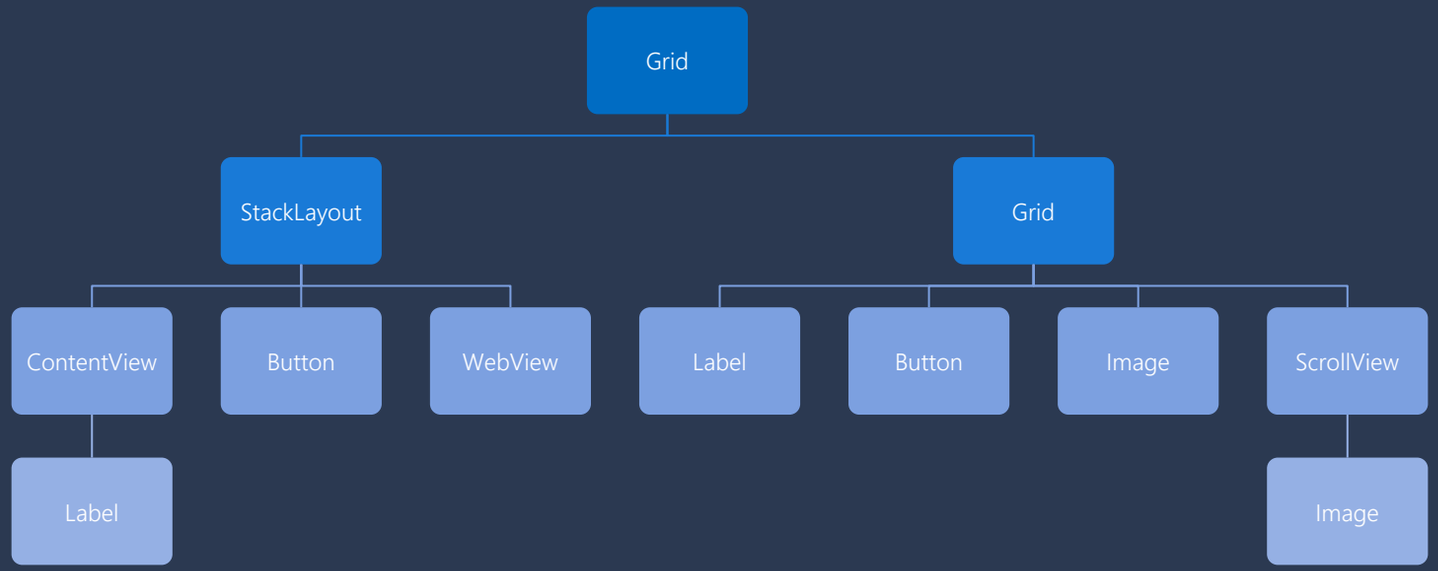

12

# Platform+Compression+Fast Renderers



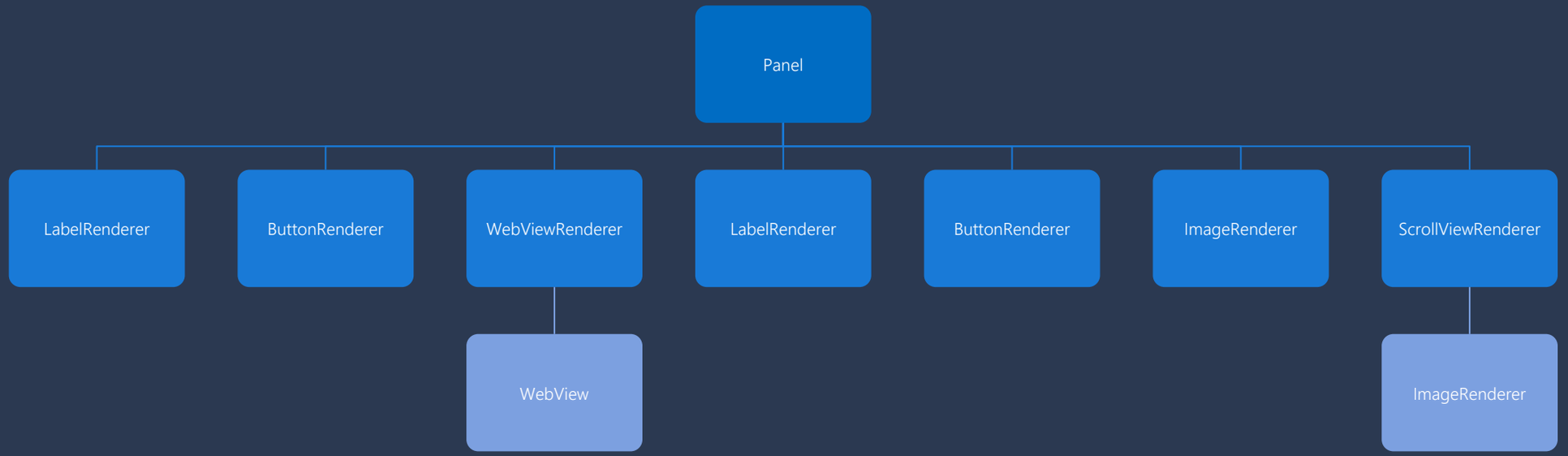

10

# Forms



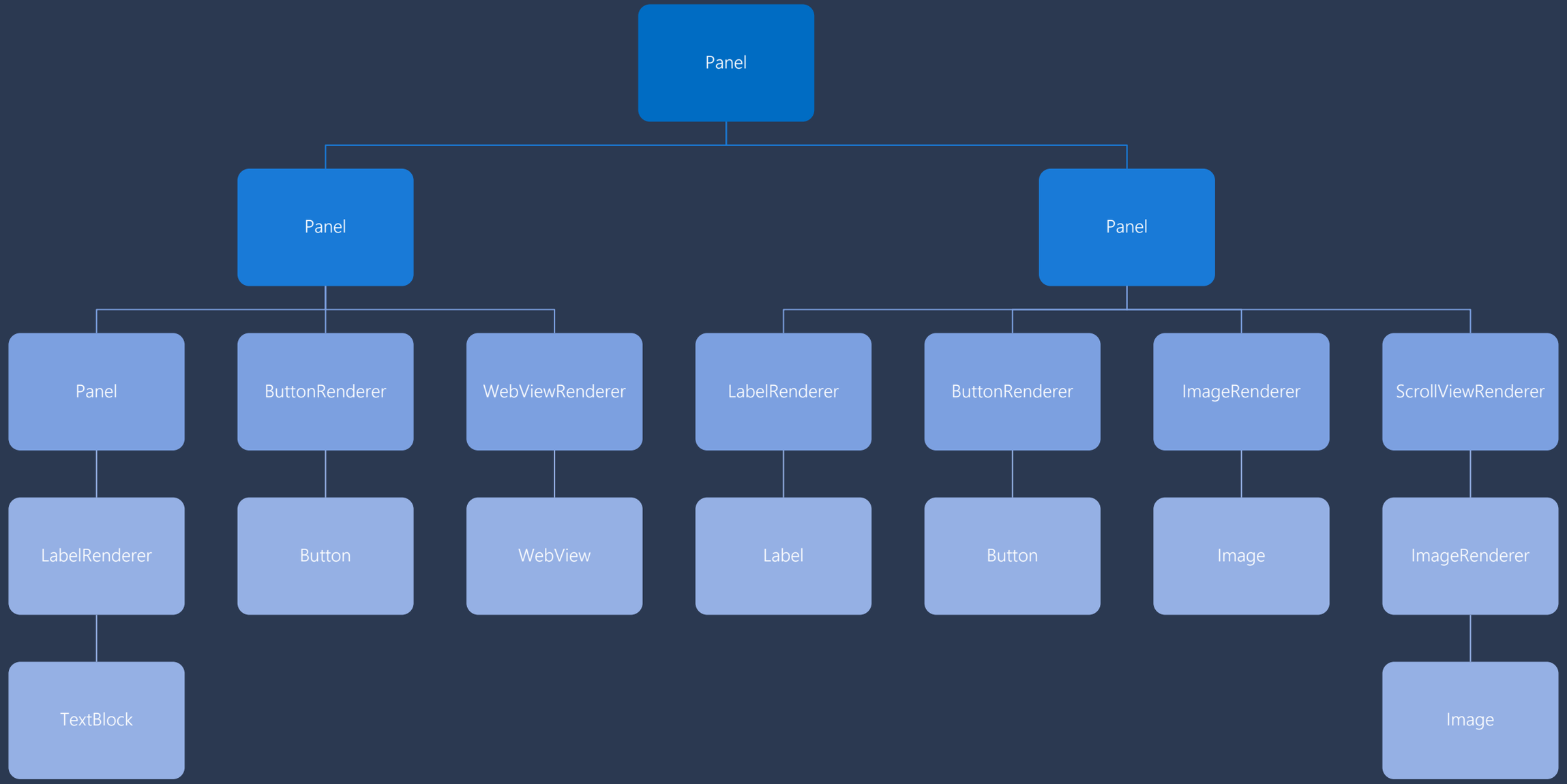

12

# Platform+Compression+Fast Renderers

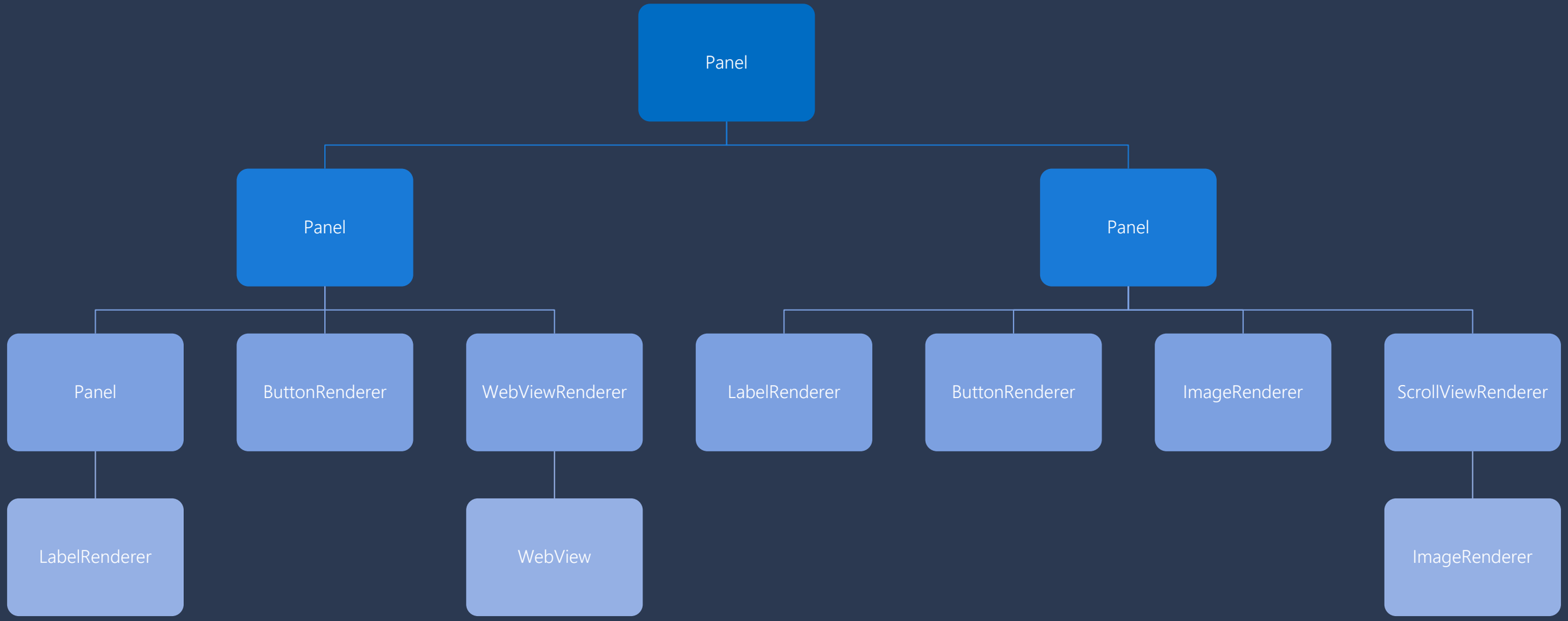



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# Forms UI Hierarchy

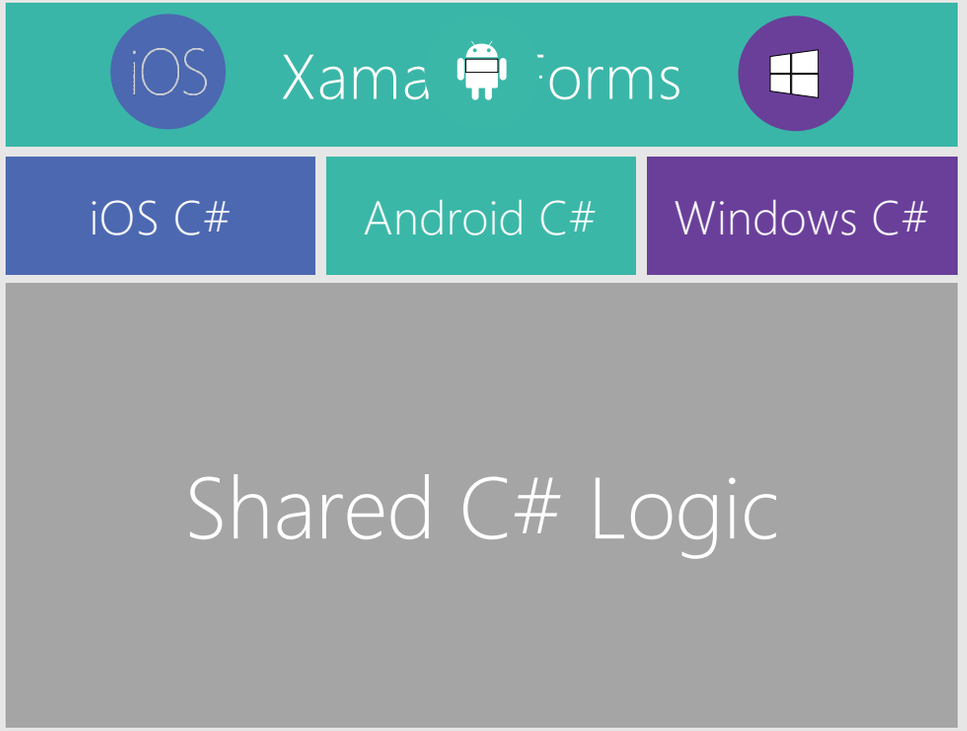


# Forms UI Hierarchy with Fast Renderers



# Page Embedding

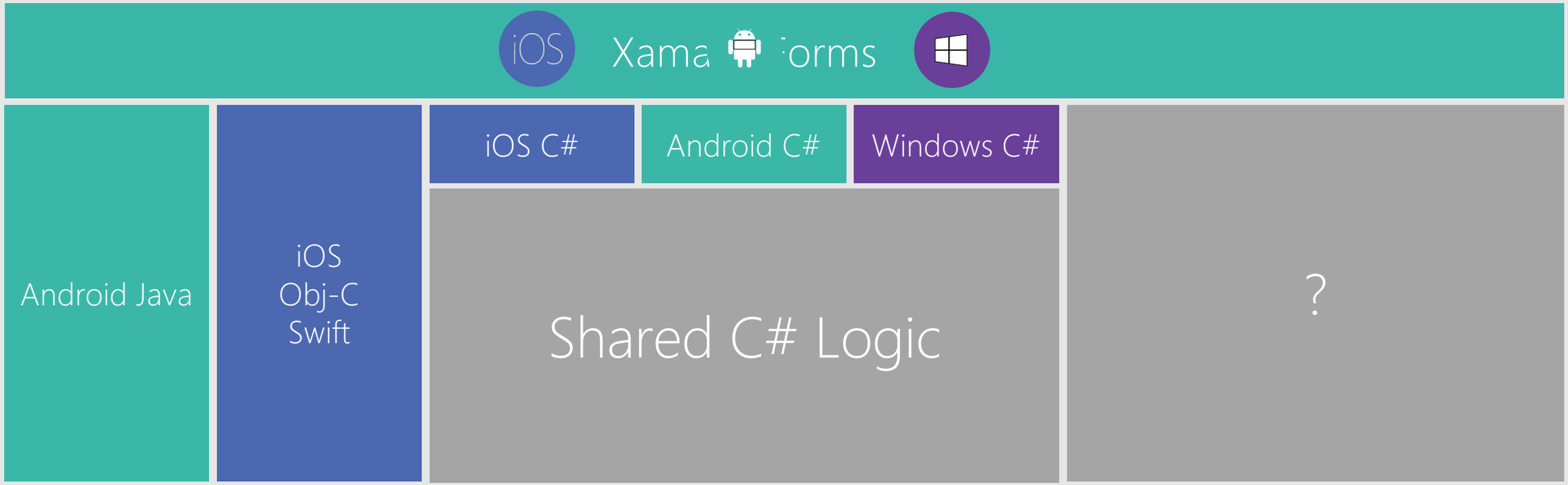
# Page Embedding



# Demo

Page Embedding

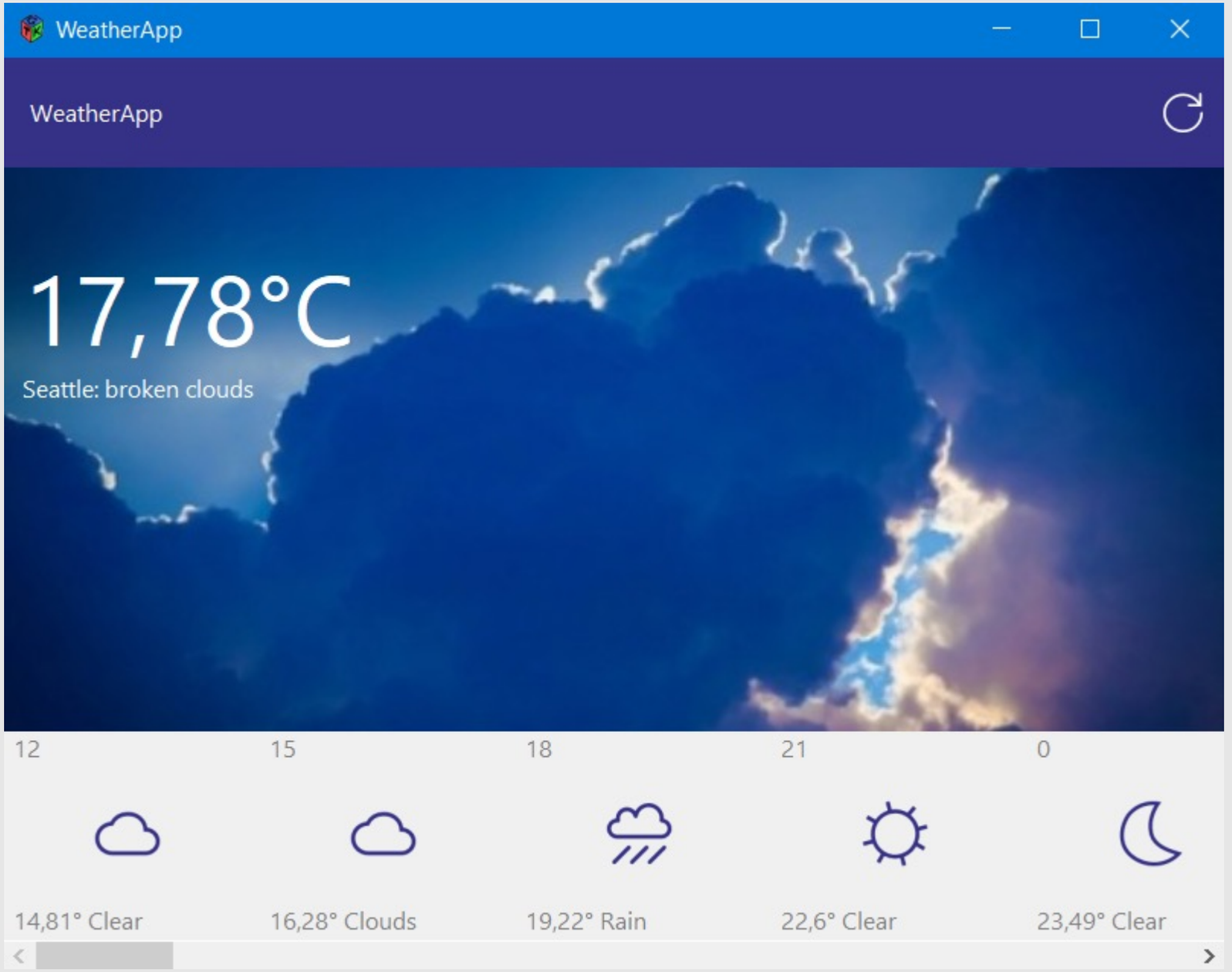
# .NET Embedding



# XAML Standard

- Lots of feedback!
- Reconciling the APIs we can without breakage
- Considering sharing implementations of
  - x:Bind
  - Visual State Manager
  - More

# New Platforms


WeatherApp
WeatherApp
17,78°C
Seattle: broken clouds
12
15
18
21
0
14,81° Clear
16,28° Clouds
19,22° Rain
22,6° Clear
23,49° Clear


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# Call to action

- Get started: xamarin.com/download
- Use Xamarin.Forms 2.4.0 today!
- Follow the very latest on our Nightly Feed:
  - github.com/xamarin/Xamarin.Forms/wiki/Nightly-Builds